Bitcoin's Required Return

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Abstract

What return should Bitcoin earn from here? In average annual terms, its price trend implies 32%, the CAPM 11–16%, holders' stated expectations 14–22%. Our MCM equation gives what holders should require at each date: market risk plus two risks the CAPM omits, concentration (holders keep much wealth in it) and mortality, measured on 520 cryptoasset deaths. This required return is consistent with monthly returns since 2012; since 2018 the two differ by less than one standard error. Today Bitcoin's required return is 18% (90% range 15–22%), about 9.5% compounded for a buy-and-hold investor, and should keep falling as holders diversify.

Keywords: Bitcoin; cryptoassets; required return; MCM equation; expected return; survival analysis; limited participation; cost of capital; portfolio construction.

1. INTRODUCTION

Suppose you must write down one number: the return Bitcoin should be expected to earn, per year, from here. Three candidate answers are on the table, and they disagree by a factor of three.

The first comes from Bitcoin's own history. A simple rule divides the exponent of Bitcoin's power-law price trend (Burger 2019) by Bitcoin's age in years. Among the candidates tested it came closest to an unbiased forecast of the returns that followed: over rolling three- to five-year periods since 2012 its forecast errors averaged nearest to zero, with the exponent re-estimated each month from the data then available (Papadogiannis 2026a). That exponent has been 5.6–6.0 since 2016; at 5.62, the value used in this paper, the rule says about 32% a year today.[1] The second comes from standard asset pricing. The CAPM asks what Bitcoin's risk would cost a diversified investor today: the ten-year Treasury yield, 5% in autumn 2026, plus Bitcoin's beta to the stock market times a forward-looking equity risk premium (ERP) of 4% (Damodaran 2026; 6% as a high case; Table 3). Bitcoin's beta to the S&P 500, estimated on month-end total returns to May 2026, is 1.75 over the last four years, 1.5 since 2016, and 1.8 since 2012 (Appendix A), which gives

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[1]The three figures in this paragraph are arithmetic average annual returns. The rule of thumb, exponent over age (β/A), gives 32% at Bitcoin's age in May 2026 ($A = 17.4$ years). Its exact five-year form, $(1+5/A)^{\beta/5}-1$, also gives 32% from September 2026 ($A = 17.7$) and 33% from mid-2026, the date of the last row of our tables. The compound (geometric) counterpart is lower by roughly half the variance, $\sigma^2/2$, and so depends on the volatility assumed: a drag of about 4.5 points at 30% volatility and 10 points at 45%, which is why the compound figure is a range, 22–28%, and the arithmetic figure a single number.

11–16% across those windows and the two premia. The third comes from asking people. In a large US household survey, those who hold crypto said they expected 22% over the next year in 2021 and 13.8% in 2025 (Candia, Coibion, Gorodnichenko, and Weber 2026); hence 14–22%.

These are not three estimates of one quantity; the disagreement begins with using one word for three things. The first is an *expected return*: a property of Bitcoin's price process, checkable only against what it later earns. The second is a *normative required return*: what an investor holding the CAPM's diversified portfolio ought to demand for Bitcoin's risk, whether or not any such investor holds it. The third is a *stated required return*: what the people who hold Bitcoin say they expect, an empirical fact about them. What is missing is a fourth, the one this paper builds: the *normative required return of the actual holder*, what the people who hold Bitcoin ought to demand given the risks their own portfolios carry. It is the one the price must satisfy in equilibrium, because they are the ones who must be willing to keep holding. That leaves a puzzle. For fourteen years Bitcoin delivered far more than the CAPM says it should. Underpricing that persistent is hard to defend, so the useful question is which risks its holders were paid for that the CAPM omits.

This paper answers by measuring those risks. Bitcoin pays no dividend or coupon, so its price must be expected to rise by enough to pay holders for the risks they take. We price three risks.

The first is *market risk*, the risk Bitcoin shares with the stock market, and the one risk the CAPM prices (Sharpe 1964; Lintner 1965). Bitcoin need not belong to a stock index for this to apply: the index stands in for the rest of an investor's wealth, and what is priced is how Bitcoin moves with it. Its beta to the S&P 500 is now about 1.75, worth about 7 points a year (throughout, returns are nominal US-dollar returns and "points" are percentage points of arithmetic average annual return), and behind it is a correlation that rose from about 0.2 before 2020 to about 0.5 since, as investors who also own equities entered (Copestake, Furceri, and Terracciano 2024; Appendix A). The second is *concentration risk*. The CAPM assumes every investor is fully diversified, so that an asset's own volatility costs nothing. Bitcoin's holders are not diversified: among US households that owned any crypto in 2021, crypto made up 19% of financial assets on average, and one in five kept more than half its financial assets in it, with similar shares reported for 2025 (Candia, Coibion, Gorodnichenko, and Weber 2026). For such a holder Bitcoin's volatility is a cost, and a larger one the larger the share of her wealth she keeps in it, which is how Merton (1987) prices assets held by few. The third is *mortality risk*, on which the CAPM is silent: the chance that the asset dies outright, through a ban, a protocol failure, an attack that breaks its cryptography, as a quantum computer might, or a collapse of the kind that ended Terra and FTX. Of the 1,815 cryptoassets in our census, 520 have died (Section 3), and the price of that risk is the standard compensation for a jump to ruin (Merton 1976).

Add the three to the risk-free rate and you have the return the holder requires. Take away the last two and what is left is the CAPM. These are the risks we judge important and can measure; others may exist (Section 8). The first two are measurable from Bitcoin's own prices and from surveys of who holds it. The third cannot be measured from Bitcoin, which has not died; it can only be measured from the deaths of other cryptoassets, as a life insurer prices one person from mortality tables of many.

Mortality risk turns out to be small for an asset like Bitcoin. We estimate mortality curves for the asset class from the census. (Earlier descriptive studies of "coin death" exist, notably Gandal, Hamrick, Moore, and Vasek (2021) and Grobys and Sapkota (2020); new here is a curated census built for pricing: death types, dates, recovery fractions, hazards by size, and the Lindy test.) The popular "Lindy" intuition,

that the longer an asset has survived the safer it is, is rejected: among tracked assets the death rate *rises* with age. And the hazard that matters for Bitcoin, the death rate of assets that reached the top twenty, measured over the whole census (2010–2026), is 3.5% a year by any cause and 0.30% a year for sudden, value-destroying collapses (Terra's LUNA and UST and FTX's FTT, all in 2022). That is worth 0.3–3.5 points of required return. Today the age rule's 32% is 27 points above the 5% risk-free rate, and the risk of death can account for only a small part of that excess.

Concentration risk was the large term, and it has shrunk. Before the ETF era crypto markets were segmented from mainstream capital (Makarov and Schoar 2020), and the holders of most of the supply were plausibly far more concentrated than the survey average. Option prices put a floor under what a fully invested holder requires (Martin 2017): 66–84% a year above the risk-free rate in 2017–22. At the 120% volatility of the early years the same holder required more than 100%.[2] Delivered returns were of that order. As volatility fell and holders became less concentrated the required return came down, term by measurable term, to about 18% today (90% range 15–22%). The sum, equation (1) of Section 2, is Merton's (1987) limited-participation pricing applied with observable inputs, not a factor regression or a consumption-based model, which cannot see who holds the asset.

Tested month by month, the required return is in line with what Bitcoin delivered. At every month-end since 2012 we compute the required return from the inputs known on that date. Like an interest rate, it is quoted per year but applies from that date on: a required return of 24% a year means about 2% over the month that follows. We set that monthly figure beside what Bitcoin returned in the same month: 172 pairs, none overlapping. We then average the required and the delivered monthly returns over the same months and multiply by twelve, so that both are quoted per year. At each date the required return is a band rather than a point, mainly because holders' combined wealth is not observed before 2021: the lower edge counts every owner, with a modest average position, and the upper edge the early holders alone, who were fully invested while the coins were in few hands. Over 2018–26 Bitcoin delivered 46% a year on average, at the top of a required-return band of 16–46% whose middle was 31%; the gap to the middle is within the noise of so volatile an asset. Over 2012–17, when the required-return band was very wide (25–202%), Bitcoin delivered 219%, twice the middle and just above the top. The test has limits (Section 5): fourteen years of a volatile asset measure an average only roughly, and the required return says nothing about the timing of booms and busts. On the same test the age rule does about as well. What separates the two is construction: the required return is built from risks rather than extrapolated from the price trend. On this reading the age rule is a reduced form of a required return that was itself falling; it now sits above the required-return band.

What this means in practice. On today's inputs, Bitcoin's price has to rise by about 18% in an average year for its current holders, taken together, to be willing to keep it (90% range 15–22%). A buy-and-hold investor compounds at less than the average, because gains and losses of equal size do not cancel; at Bitcoin's current volatility the drag is about 9 points, so 18.3% compounds at about 9.5% a year. The 18% is the figure to compare with other expected returns, the 9.5% the rate at which wealth would grow. Both are today's figures and move with the inputs.

For an allocator three things follow. First, the age rule's 32%, an extrapolation of Bitcoin's price trend, is 14 points above what its holders now require. Second, Bitcoin is not a free diversifier: about 7 of the

[2]The concentration term is risk aversion times variance times portfolio weight, $\gamma\sigma^2 w$ (Section 2). At $w = 1$, $\gamma = 1$ and 120% volatility it is $1.2^2 = 1.44$, or 144 points a year above the risk-free rate. Today, with γ of 1–2.8, $\sigma = 42\%$ and w of 8–19%, the same term is 1–9 points.

18 points are payment for stock-market risk the allocator already holds. A diversified investor making a first purchase would bear that market risk and a small risk of death but no concentration risk, so she should require less than its present holders do: the 5% risk-free rate, plus 6–8 points for its beta, plus 1–2 for death, or 12–15%. Third, the 18% tells her how much to hold: each 1% of her wealth in Bitcoin raises the return she needs by about half a point, so she stops at about 11% of financial wealth on the central numbers (Section 7).

Contributions and plan. The paper makes five contributions. The first is the MCM equation: the CAPM extended by the two risks it assumes away, written for Bitcoin's actual holders (Section 2). The second is a measurement of concentration, the largest term through most of Bitcoin's history. We define the holder weight as Bitcoin's capitalization over its holders' wealth, anchor it on surveyed portfolios, and carry it through time for the average owner and for the early holders. Concentration supplied about 87% of the required return in 2013 and supplies about a third today (Sections 4 and 5, Appendix B). The third is a census of cryptoasset deaths built for pricing, in which the Lindy hypothesis is rejected and the hazard of top assets is measured; it shows that mortality was never the large term (Section 3, Appendix C). The fourth is a month-by-month test of the required return against Bitcoin's returns since 2012 (Section 5). The fifth places Bitcoin among the assets an allocator already holds, on the hypothesis that it is priced at what its holders require, so that the required return is also the return to expect. Four equity indices, gold and Ether are priced on the same inputs, which shows that Bitcoin's compound required return, about 9.5% a year, is close to that of equities at three times the volatility; a hurdle and a sizing rule for a diversified investor follow (Section 7). The equation is applied to Ether and smaller cryptoassets (Section 6) and to gold (Appendix E), where it reproduces a known result.

2. THE REQUIRED-RETURN EQUATION

Expected versus required. An asset's *expected return* is a forecast: what it will earn on average. Only realized returns can check it, and they check it badly: Elton (1999) shows that realized returns are so noisy that decades of data can fail to distinguish competing values. A *required return* is a hurdle: what an investor needs the asset to earn to be willing to hold it, given its risk. No market quotes a required return. A bond's can be read from its yield, and a stock's can be backed out of its price and expected cash flows; Bitcoin has no cash flows, so that route is closed. What its holders require can be read from what they say (Greenwood and Shleifer 2014), from the portfolios they choose (Giglio, Maggiori, Stroebel, and Utkus 2021), and from option prices (Martin 2017), or it can be derived from a model of what an investor ought to demand. The CAPM is the familiar model of that kind (Sharpe 1964; Lintner 1965), but it gives the hurdle of a fully diversified investor, and Bitcoin's holders are not diversified. This paper derives the hurdle for the holder Bitcoin actually has, with her concentration, and checks it against what holders say and what options imply (Section 4) and against what the asset delivered (Section 5). The last check rests on equilibrium: the price adjusts until the expected return equals what its holders require. That equality is what "the market is in equilibrium" means; it is why Cochrane (2011) can treat expected returns and discount rates as one object, and testing it is what empirical asset pricing does (Fama and French 2004). Section 7 uses the same equality the other way: if Bitcoin is priced in equilibrium, today's required return is also the return to expect. For an asset whose holders are still learning what it is, the gap can stay open for years, because learning makes realized returns predictable in ways no required return explains (Lewellen and Shanken 2002; Pástor and

Veronesi 2009a).

Whose requirement? A required return is always someone's. A pension fund might be happy to hold Bitcoin at an expected 8%, but if pension funds do not hold it, their 8% has no bearing on the price. The price is set by the people who do hold the asset: it must offer enough for them to go on holding it. That is what this paper means by Bitcoin's required return. It is a property of the asset and its holder base, not of one investor, and it changes when the holders change.

What risk aversion means. A risk-averse person would pay to avoid a risk, even one with fair odds. Risk aversion, γ (the Arrow–Pratt coefficient), says how much: to avoid a risk with variance v, a person with risk aversion γ would give up about $\gamma/2 \times v$ (Pratt 1964; Arrow 1971). Both the risk and the payment are measured as shares of her portfolio. Take a coin toss for 10% of it. The variance is $0.10^2 = 0.01$, so a person with $\gamma = 2$ would give up $2/2 \times 0.01 = 1\%$ of her portfolio to avoid it, and a person with $\gamma = 1$ half a percent. ($\gamma = 1$ is the benchmark economists call log utility.) Estimates land between 1 and 4: about 2 from US household portfolios (Friend and Blume 1975), below 2 from labor-supply behavior (Chetty 2006), a median of 1.7 in Italian household panels (Chiappori and Paiella 2011), and small single digits in the macro calibrations that followed Mehra and Prescott (1985); 2–3 is the workhorse range.

What concentration costs. Holding a volatile asset is a risk of the same kind. Suppose an investor keeps a share w of her portfolio in an asset whose volatility is σ a year (for now, one that moves independently of the rest of her portfolio). Her portfolio then moves by w times what the asset does, so the risk she bears has variance $\sigma^2 w^2$, and by the rule above she would give up $\frac{\gamma}{2}\sigma^2 w^2$ a year to be rid of it. The position earns her $w(\mu - r_f)$: her share times the asset's expected return μ above the risk-free rate r_f, its *excess return*. She chooses the share that makes the difference, $w(\mu - r_f) - \frac{\gamma}{2}\sigma^2 w^2$, as large as possible. At that share its derivative with respect to w is zero, $\mu - r_f - \gamma\sigma^2 w = 0$, so that $\mu - r_f = \gamma\sigma^2 w$ (Merton 1969): one more dollar in the asset would earn exactly what its extra risk costs. The excess return she requires is therefore her risk aversion times the asset's variance times her share. With Bitcoin's current volatility, $\sigma = 0.42$ ($\sigma^2 \approx 0.176$), $\gamma = 2$ and the surveyed share of 2021, 19% (Section 4), that is $2 \times 0.176 \times 0.19 \approx 6.7$ points a year above the risk-free rate; at a share of 5% it is 1.8 points. The three inputs multiply, so halving the share lowers the requirement as much as halving risk aversion or the variance would. A broader holder base lowers the requirement only if it lowers the share, which needs holders' wealth to grow faster than the asset's capitalization. The ETF era did that, on the evidence available: institutions that report Bitcoin ETF positions hold under 1% of their portfolios in them on average (CoinShares 2025), against the 19% of surveyed owners. Read from left to right this is a sizing rule: given an expected return, it gives the share to hold (Section 7). Here it is read the other way. If holders hold the share that is best for them, an observed share and an observed volatility reveal the excess return the holder must have been requiring, which is how Merton (1987) prices an asset held by few. Adding the two risks this simplest version leaves out, market risk and sudden death, gives the paper's central equation, the **required-return equation**:

$$\mu - r_f \;=\; \underbrace{\beta\,\mathrm{ERP}}_{\text{market risk}} \;+\; \underbrace{\gamma\sigma^2 w}_{\text{concentration risk}} \;+\; \underbrace{\lambda\,(1-\delta)}_{\text{mortality risk}} \tag{1}$$

We call it the *MCM equation*, for market, concentration and mortality. On the left is the asset's expected return over the risk-free rate; on the right are the three risks, each stated per dollar held in the asset. β is Bitcoin's beta to the equity market and ERP the equity risk premium. The **concentration term** is γ times

variance times the holder's own portfolio weight, the risk she cannot diversify because she chooses, or is forced, to hold a lot of one thing. Every input carries a date, so the equation gives the required return at a point in time, not a constant of the asset.

The mortality term. A holder needs compensation for the risk that the asset dies, equal to the loss she can expect from it. That expected loss is the product of two numbers, as in the pricing of jumps and of default (Merton 1976; Duffie and Singleton 1999). λ is the probability per year that the asset dies. δ is the *recovery fraction*, the share of her money a holder keeps when it does, so $(1-\delta)$ is the share she loses. They are multiplied because dying and losing everything are not the same thing: when TerraUSD collapsed, holders kept about two cents on the dollar ($\delta \approx 0.02$; Liu, Makarov, and Schoar 2023); when BUSD was wound down under regulatory order, holders redeemed at par ($\delta \approx 1$; NYDFS 2023). A 3% annual death probability with 95% lost on death is an expected loss of $0.03 \times 0.95 \approx 2.9$ points a year, and that is what the price must add, in the years the asset lives, to what holders otherwise require. A surviving asset's history contains this compensation in full, because the losses it was paid for never arrived (Brown, Goetzmann, and Ross 1995). The required return of equation (1) contains it too, which is what makes the two comparable.

Where the equation comes from. Each term of equation (1) rests on an established result. The concentration term is textbook portfolio choice (Merton 1969) used as a pricing statement under limited participation (Merton 1987): when only some investors hold an asset, they must be paid for the idiosyncratic variance they absorb. The mortality term is the standard jump compensation (Merton 1976) read through the survival conditioning of Brown, Goetzmann, and Ross (1995), the form, hazard times loss, used in event-risk allocation (Liu, Longstaff, and Pan 2003; Das and Uppal 2004) and in the rare-disaster models that followed Barro (2006). The beta term is the CAPM. The equation reduces to it under two conditions: the hazard λ is zero, and every holder's weight w equals Bitcoin's share of all investable wealth, under 1%. The concentration term is then below 0.2 points and is no longer an addition to the CAPM: it is Bitcoin's own contribution to the risk of the market portfolio, which the CAPM counts when its market includes Bitcoin. What is new is the assembly and the measurement. Equation (1) joins the three terms into one pricing statement, measures each on its own data, and checks the result against holders' stated expectations, option prices and fourteen years of returns. Required returns for Bitcoin have been read from option prices (Almeida, Grith, Miftachov, and Wang 2024) and from factor regressions (Liu and Tsyvinski 2021). To our knowledge none has been built from the holders' side. The nearest work runs the other way, from investors' beliefs to their demand or their weights (Benetton and Compiani 2024; Duchin, Solomon, Tu, and Wang 2025); Appendix E.3 compares other estimates. The concentration mechanism is Merton's (1987) *investor recognition hypothesis*, which has an empirical record: assets held by few carry a higher cost of capital, which falls as recognition broadens. Levy (1978) and Errunza and Losq (1985) derived the same form for investors who hold few securities and for securities that foreigners cannot hold. The hypothesis has been tested for individual stocks, where broadening the shareholder base raises the price and lowers subsequent returns (Kadlec and McConnell 1994; Lehavy and Sloan 2008; Bodnaruk and Östberg 2009), and for national markets opening to foreign capital (Bekaert and Harvey 2000). Pástor and Veronesi (2009b) give it a lifecycle: a new technology's risk is idiosyncratic while it is small and held by few, and turns systematic as it is adopted. Ibbotson, Idzorek, Kaplan, and Xiong (2018) generalize it into a theory of popularity, in which hard-to-diversify assets earn a premium that popularity erodes. For cryptoassets the record is indirect: a broadening investor base is linked to idiosyncratic volatility (Izadyar and Zamani 2022), and the participa-

tion of institutional investors explains the changes in crypto's correlation with equities (Copestake, Furceri, and Terracciano 2024), as index investment did for commodities (Tang and Xiong 2012). Every test above is a sign test on prices or correlations; this paper estimates the level.

Four clarifications. First, on what the equation explains. A full model of Bitcoin's price would start from the supply schedule, the pace of adoption and the costs of miners, and derive the price, its volatility, and who holds the asset (Biais, Bisière, Bouvard, Casamatta, and Menkveld 2023; Pagnotta 2022; Cong, Li, and Wang 2021). Such models take the rate at which investors discount Bitcoin's future as given, or leave it to be estimated. That rate is the object of this paper. Accounts that locate a blockchain's value in its ecosystem of firms and users (Papadogiannis 2026b) are complementary: they say where value comes from, this paper at what rate it is discounted. We take volatility, the holder's weight and the death hazard as observed. Second, the equation is special neither to Bitcoin nor to one holder. The same line prices founder stock and private equity (Moskowitz and Vissing-Jørgensen 2002; Kahl, Liu, and Longstaff 2003; Kerins, Smith, and Smith 2004) and emerging-market equities before liberalization; Bitcoin enters through the values of the inputs, and through one data advantage: a thousand-asset graveyard from which λ can be measured rather than assumed. Every holder satisfies the equation with her own weight and risk aversion, since the more risk-averse hold less. Their holdings must add up to the whole asset, so the premium that clears the market is the same equation with the holders' combined weight (equation 2 in Section 5) and a wealth-weighted average of their risk aversions (Appendix B). No one holder sets the price; all do, in proportion to wealth. Emerging markets show the mechanism we rely on (Bekaert and Harvey 2000; Henry 2000): required returns are high while a market is segmented and fall when it opens to outside capital; the ETF era is Bitcoin's liberalization. Third, on using the survey average for w. The equation needs the combined weight, and the survey reports the average of household weights. The two coincide when holders have similar risk aversion, since each then adds Bitcoin until $\gamma\sigma^2 w$ reaches the same premium; otherwise the average is a bound (Section 4). The reading also assumes that holders chose their weights on beliefs that are right on average. A weight inflated by over-optimism satisfies the same equation but reveals a belief, not a fair premium, and only the comparison with realized returns in Section 5 can tell the two apart. Fourth, on whom the equation describes: an investor whose risk aversion does not change with her wealth, who faces two kinds of risk in Bitcoin, ordinary price swings and the chance of sudden death, and who keeps the rest of her wealth in the broad market (Appendix A).

What the equation leaves out. Equation (1) does not price everything, but what it omits has known signs. Pushing the required return up: a liquidity premium (Amihud and Mendelson 1986), material in Bitcoin's early years and small today; a jump-risk premium above the actuarial loss if deaths are systematic, which Section 7 charges for; ambiguity about the distribution itself, natural for a new asset; hedging demands against state variables such as monetary policy, to which Bitcoin has become sensitive (Copestake, Furceri, and Terracciano 2024). Pushing it down: a convenience or service yield from holding the asset, the analogue of gold's unobserved dividend in Barro and Misra (2016); a lottery preference for skewness (Barberis and Huang 2008); and disagreement under short-sale constraints, which raises prices and lowers expected returns (Miller 1977). Their net effect cannot be measured, only bounded. Option prices set a ceiling: a holder with *all* her wealth in Bitcoin requires 21–35% a year today, whatever risks she prices (Section 4), and today's holders, with 8–19% of their wealth in crypto (19% in the 2021 survey; 8% once fund and treasury investors are counted, Section 4), must require less. Holders' stated expectations, 22% in 2021 and 14% in 2025,

lie in the lower part of the equation's range for those dates or below it, which does not suggest a missing premium. Realized returns cannot decide the matter: since 2018 they agree with the required return within fifteen points a year, with a standard error of 24 (Section 5). A net omission of a few points in either direction cannot be excluded, and it would move the 2026 estimate one for one.

What moves each term. Volatility falls as markets deepen (Bitcoin's from about 150% in 2013 to 40–50% in 2026). The weight w falls when holders' wealth grows faster than the asset's capitalization: when the only people willing to hold an asset are enthusiasts, each must hold a lot of it; when ETFs let everyone hold a sliver, nobody has to. The market term moves the other way, for the same reason. Beta measures how Bitcoin moves with the rest of its holders' wealth. Once holders also own equities, they sell both when their wealth or their appetite for risk falls. This happened in 2020, four years before the spot ETFs (Appendix A.3). The mortality term is the one with no trend. The census finds that the hazard of cryptoassets does not decline with age (Section 3), so we hold it constant, and new threats such as quantum computing could raise it (Appendix C). Figure 2 in Section 5 shows the path of each.

3. DATA, AND THE CENSUS OF CRYPTOASSET DEATHS

Market risk and the risk-free rate. Beta is estimated against the S&P 500, on month-end total returns of the Vanguard S&P 500 ETF (Yahoo Finance) over four-year windows. We use this index because the surveyed holders are US households and the ERP is measured on it; a global index gives nearly the same beta (Appendix A). The premium is held at 4%, close to the 4.2% implied by the S&P 500 at the start of 2026 (Damodaran 2026). The risk-free rate is the ten-year Treasury yield. Table 7 and Appendix E give the sources for other equity indices and gold.

Concentration risk: prices, surveys and options. Volatility comes from Bitcoin's monthly prices (Coin Metrics 2026). The holder weight comes from household surveys: crypto-ownership rates from the Federal Reserve's payment and household surveys and Pew (0.4% of US adults in 2014, 1.2% in 2018, 11% in 2021, 12% since), and owners' portfolio shares and expected returns from Candia, Coibion, Gorodnichenko, and Weber (2026), with Austrian household-finance data (OeNB 2025) as a European cross-check. Carrying the weight through time also uses Bitcoin's market capitalization and address counts (Coin Metrics), global household wealth (UBS Global Wealth Report), and spot-ETF and public-company holdings (Bitcoin Treasuries 2026); Appendix B shows each step. Risk aversion comes from the household-finance literature and from surveys of crypto owners' risk tolerance (Section 4). The requirement of a fully invested holder comes from Deribit option prices and the DVOL index.

Mortality risk: a census of cryptoassets. The asset panel is the Coin Metrics community dataset, observed July 2010 to May 24, 2026. After removing fiat exchange rates, tokenized equities, wrapped and staked duplicates, stablecoins, and trading-pair artifacts, the universe is **1,815 cryptoassets** and about 67,700 asset-months of exposure. Daily trading volume is available for nearly all and daily prices for the 139 historically significant ones, so volume dates births and deaths and prices give returns. We date an asset's birth to the first day it shows a price or volume. We count it as dead when its data feed ends more than 180 days before the dataset does (234 cases), when its price falls below 1% of its all-time high and never recovers (33 cases, on the price-complete subset), or when its trailing three-month volume falls below 1% of its peak and below $250,000 a day and never recovers (438 cases, overlapping the other two). The volume rule matches the abandonment criterion of Gandal, Hamrick, Moore, and Vasek (2021), with a floor

that spares mid-caps whose volume merely normalized after 2021; the other thresholds are conventions (Fantazzini 2022 reviews the alternatives), and the age result holds under narrower and broader definitions (Appendix C). Ticker migrations are not deaths: when MATIC became POL or EOS became Vaulta, holders received successor tokens one for one (Polygon Labs 2024; Binance 2025), and seventeen such events are removed. Catastrophes are dated from the public record: LUNA and UST at May 12, 2022, when Terra's blockchain was halted (Melinek 2022), and FTT at November 10, 2022, when FTX's assets were frozen (Securities Commission of The Bahamas 2022). The result is **520 deaths among 1,815 assets (29%)**, of which 8 are sudden, value-destroying *jump-collapses* and the rest gradual abandonment. Coin Metrics adds an asset only once it is significant, so micro-caps that died young never enter the panel, the instant-history bias of fund databases (Fung and Hsieh 2000). The panel therefore *understates* the death rate at young ages, which studies of all listed coins find to be high (Lansky 2020; Aldhahi and Alsamaani 2026). Our results on age hold for tracked assets (Appendix C.4). The elite estimates used for Bitcoin are unaffected: an asset in the top twenty is fully observed thereafter.

Table 1: Death rates of cryptoassets by size.

Group	Assets	Deaths	Death rate a year
All assets, in quarters by peak trading volume			
Smallest quarter	452	147	17.4%
Second quarter	451	160	13.0%
Third quarter	451	145	9.8%
Largest quarter	452	67	3.2%
Ever in the top 20 by volume	**219**	**36**	**3.5% (2.5–4.9%)**
of which sudden collapses	**219**	**3**	**0.30% (0.06–0.86%)**

Deaths per asset-year observed; 95% confidence intervals in parentheses. Rows in bold: the upper and lower ends of Bitcoin's mortality term.

What the census shows. Two results matter for pricing (Appendix C). First, tracked cryptoassets do not become safer with age. The popular "Lindy" idea, that the longer an asset has survived the longer it will survive, would make the death rate fall like one over age. The data reject it. The death rate rises from under 5% a year in an asset's first year to 12–17% between ages three and nine, and a regression of deaths on age with calendar-year effects gives an elasticity of **+0.42 (s.e. 0.04)** where Lindy requires -1. Cryptoassets age as young firms do, whose risk of failure also rises over the first years of life before it falls (Brüderl and Schüssler 1990); Papadogiannis (2025, 2026b) draws the wider parallel with firms. A digital asset gives a further reason to doubt Lindy, one the census cannot show: its cryptography is fixed at design, while the means of attacking it only improve (Appendix C.5). Second, size is what protects (Table 1): the death rate falls from 17.4% a year in the smallest quarter by trading volume to 3.2% in the largest. Of the 219 assets that ever reached the top twenty by monthly volume, 36 died over 1,017 asset-years, 3.5% a year, and only three were sudden collapses: LUNA, UST and FTT, each a near-total loss within days. None has occurred since. The others were slow fades.

The mortality term used for Bitcoin. The census gives a low and a high reading of the term, and we use the range between them, **0.3 to 3.5 points a year**, at every date, since hazard shows no decline with age. The low reading counts sudden collapses only, the event the term is defined for: Table 1 gives 0.30% a year for λ, and holders kept one to five cents on the dollar (Appendix C.3), so $\lambda(1-\delta) \approx 0.3$ points. The

high reading counts every top-twenty death as a sudden total loss: 3.5% a year in Table 1 (2.4% without the twelve deaths of 2025, mostly delistings). It overstates the term: in a slow fade holders can sell on the way down, and that loss is already in the asset's volatility and average return. Option prices agree, capping the probability of a collapse to zero at about 1.4% a year (Appendix C.5). The truth lies between the two, and central estimates use the middle, 1.9 points. With the low reading alone the 18% of Section 4 would be 16.6%; with the high reading alone, 19.8% (Appendix D). Risks that have never struck any asset, notably a quantum break of Bitcoin's signatures, would raise the term but cannot be priced from the census (Appendix C).

4. WHAT BITCOIN'S HOLDERS REQUIRE

The required-return equation, equation (1), needs five inputs. The graveyard supplied the mortality term. This section supplies the holder's portfolio weight and risk aversion, takes volatility from prices and the market beta from Appendix A, assembles the required return for 2026, and sets it against two outside readings, holders' stated expectations and option prices.

Source 1: holders' measured portfolios. The Cleveland Fed study of cryptocurrencies in household finance (Candia, Coibion, Gorodnichenko, and Weber 2026) reports the number the equation needs most. Among US households that owned any crypto in 2021Q3, crypto was **19% of financial assets on average**; 40% of owners held under 5%, and about a fifth held more than *half*. The 2025 wave reports similar shares without publishing the figure. Three points govern its use.

It is an upper bound. The 19% averages household shares, each counted once. What sets the price is the share of owners' combined wealth (equation 2), which is not reported and is lower if wealthier owners hold smaller shares. A European cross-check agrees: in the Austrian central bank's 2023 household survey, owners rarely held above 20–25% of financial assets in crypto (OeNB 2025).

It measures crypto, not Bitcoin alone, which is what the equation calls for. The concentration term prices how much of the holder's portfolio moves with Bitcoin: her Bitcoin share, plus her share in each other coin times that coin's beta to Bitcoin. Those betas are 0.85–1.15 (monthly, since 2018, for Ether and five other large coins), so the total is close to her crypto share, and we use that share as Bitcoin's w. Measured directly, the beta of the whole crypto market on Bitcoin is 1.0 (Section 6).

For 2026 we use a range of 8–19%. The upper bound is the survey's 19%; the market-clearing back-cast of Section 5 gives 23% for holders of the surveyed kind, and 15% when run on all cryptoassets (Appendix D). The lower bound adds a second class of holders, whom no survey counts as "crypto owners": investors in spot ETFs and in public companies that hold Bitcoin, about 2.5 million BTC by mid-2026. They are assumed to hold 1.5% of their wealth in it, which brings the combined weight to 8% (Appendix B).

Source 2: holders' stated expectations, a partial check. The same surveys asked holders what return they expected over the next year: 22% in 2021Q3, 13.8% in 2025Q2. Table 2 sets each answer against equation (1) evaluated with the inputs of its date.

The table is not a clean confirmation. In 2021 holders' stated 22% lies inside the model's 18–38% and is reproduced at a risk aversion of about 1.3, close to log utility: beliefs and positions cohere, as Giglio, Maggiori, Stroebel, and Utkus (2021) find for equity investors. In 2025 they do not. Holders said 13.8%, barely more than the risk-free rate plus the market term alone (12.2%), which leaves almost nothing for a fifth of their wealth in an asset with 53% volatility; it would take a risk aversion of about 0.2. Either

Table 2: Two survey waves, one equation.

	2021Q3	2025Q2
Volatility σ (at the survey date)	77%	53%
Holder portfolio weight w	0.19 (measured)	0.19 ("similar across waves")
Risk-free rate	1.5%	4.4%
Market term (rolling beta $\times$ 4%)	5.2 pts ($\beta = 1.3$)	7.8 pts ($\beta = 2.0$)
Required return at $\gamma \in [1, 2.8]$	18–38%	18–27%
Stated expected return	22%	13.8%
γ that reproduces the stated return	≈ 1.3	≈ 0.2

Mortality term at 0.3 points. Volatility is the measure of Section 5 (weights halving every nine months), averaged over the quarter each wave was fielded; Candia et al. report the 2025 wave as 2025Q2 and the 2025 portfolio share as similar to 2021's. Betas: the rolling four-year estimates of Appendix A.

holders in 2025 expected less than their own risk required, or they do not perceive the market risk the rolling beta measures, or a twelve-month point forecast given after a strong run is not the long-run expectation the equation prices. The paper therefore does not calibrate risk aversion to the surveys.

The range for risk aversion. The lower end of the range for γ is below the household estimates of Section 2, because the people who choose to hold crypto are more tolerant of risk than other households. In the Federal Reserve's 2022 household survey about 40% of crypto investors rated their willingness to take financial risk at 7 or more out of 10, against under 20% of non-owners (Hayashi and Routh 2025); Austrian owners are likewise more risk-tolerant than non-owners (Stix 2021), and retail investors who buy crypto products hold risky portfolios and tilt further toward risk after their first crypto purchase (Hackethal, Hanspal, Lammer, and Rink 2022). These studies rank holders against non-holders but do not measure γ. A lower bound of 1, log utility, is our reading of them; it brackets the 1.3 implied by the 2021 survey. The choice matters. With the lower bound at 1.7, the median household of Chiappori and Paiella (2011), the central estimate below would be 19.2% instead of 18.3%, and as γ goes to zero the required return falls to 12–15%, the diversified investor's hurdle of Section 7, which is where the 2025 answer sits.

Source 3: option prices. Options answer a question the surveys cannot: what would a holder with *everything* in Bitcoin require? No holder can be more concentrated, so her requirement is a ceiling for the concentration term. Martin (2017) shows that it can be read from option prices: an investor fully invested in an asset, with log utility ($\gamma w = 1$ in equation (1)), requires an excess return equal to the asset's variance as option prices imply it, not as history shows it. For July 2017 to December 2022, when implied volatility ran at 80–90%, Deribit options put this at **66–84 points a year** (Almeida, Grith, Miftachov, and Wang 2024). Crypto markets were then segmented from mainstream capital (Makarov and Schoar 2020), so holders close to all-in are plausible for those years; Section 5 builds the upper edge of the band on such a holder. Today implied volatility (Deribit's DVOL index) has run between about 40 and 55, so the same all-in holder would require 0.40^2–$0.55^2 \approx$ **16–30 points of excess return**, or 21–35% a year in nominal terms. Today's holder is not all-in: with w of 8–19% she requires **1–9 points** for concentration, on top of the risk-free rate and the market term.

Putting the 2026 number together. Table 3 lists every input of equation (1) and its source. Volatility, beta, the risk-free rate and the death hazard are measured; the ERP is a convention; risk aversion and today's holder weight are bounded, not measured, and carry the widest ranges. On a risk-free rate of 5.0%, the market term is 6–8 points, concentration 1–9 points and mortality 0.3–3.5 points. Summing the extremes

gives an *envelope* of 13–26% a year, which puts every input at its extreme at once. We therefore treat every value in a range as equally likely and draw the four ranged inputs independently, since they come from separate sources (prices, surveys, a census), 400,000 times (Appendix D). The median of the resulting required returns is **about 18% a year**; 68% fall in 16–20% and **90% in 15–22%**. These ranges describe uncertainty about the inputs; they are not statistical confidence intervals. The market term is now the largest component (about 7 points), followed by concentration (about 4.5); the holder weight contributes most of the width. Beta's range spans point estimates, not sampling error; with that error the 90% range is 14–23% and the median is unchanged (Appendix D). This is an arithmetic average; the compound return it implies is lower by about half the variance, roughly 9 points today, so about 9.5% a year. A realized five-year return will also differ from it by roughly ±19 points a year on the compound return (one standard deviation), the noise of a 42%-volatility asset.

Table 3: The inputs to the 2026 required return, and where each comes from.

Inputs with one value			
Risk-free rate	**5.0%**	The ten-year Treasury yield in autumn 2026 (4.9–5.3% from the end of September to November 11, 2026).	
ERP	**4%**	Forward-looking: the premium implied by US stock prices was 4.2% at the start of 2026 (Damodaran 2026). Its recent high, 5.9% at the end of 2022, gives the high case of 6% (Appendix A).	
Volatility, σ	**42%**	Monthly returns to May 2026, exponentially weighted with a half-life of nine months. Option-implied volatility has run at 40–55%; Appendix D varies σ.	
Inputs with a range		**Low end**	**High end**
Beta	**1.5–2.0**	1.5: all months since 2016. Two- and three-year windows give 1.3–1.4, with larger standard errors (Appendix A).	2.0: the upper part of the four-year estimates since mid-2022 (1.6–2.2); the latest four- and five-year estimates are 1.75 and 1.8.
Risk aversion, γ	**1–2.8**	1 (log utility): crypto owners tolerate more risk than other households, and the 2021 survey implies 1.3.	2.8: the upper part of the 2–3 range common in household finance; three-quarters of households are below 3 (Chiappori and Paiella 2011).
Holder weight, w	**8–19%**	8%: Bitcoin's value over the combined wealth of surveyed owners (about $7 trillion) and of ETF and corporate-treasury investors (about $13 trillion, if their Bitcoin is 1.5% of their wealth; Appendix B).	19%: the crypto share of owners' financial assets in the Cleveland Fed survey (2021; similar in 2025).
Mortality, $\lambda(1-\delta)$	**0.3–3.5 points**	0.3: sudden collapses only, 0.30% a year at a 95% loss.	3.5: every death of a top-twenty asset, slow fades included, counted as a total loss (Table 1).

Market term: beta times the ERP. Concentration term: $\gamma\sigma^2 w$. Any value between the two ends of a range is treated as equally likely (see the text).

5. THE REQUIRED RETURN THROUGH TIME, AGAINST WHAT BITCOIN DELIVERED

A cost of capital earns its name if, over time, the asset delivers about what it says. This section checks that in three steps: compute the required return at every month-end since 2012, measure what Bitcoin delivered over the following month, and compare the two. The sample starts in January 2012 because Bitcoin's price series begins in July 2010 and the volatility estimate needs about eighteen months of it.

Step 1: the required return at each month-end. At every month-end we evaluate equation (1) with the inputs as they stood on that date. Three inputs that move with time are read directly from the data (Figure 1):

- *Volatility*: the standard deviation of monthly returns up to the date, with weights that halve every nine months back in time. There is no fixed window: the latest nine months carry half the weight and the latest three years 94% (Appendix A.7).
- *Beta*: estimated on the previous four years and capped at 3 (Appendix A).
- *The risk-free rate*: the ten-year Treasury yield of the date.

Risk aversion and the mortality term keep the ranges of Table 3 throughout.

The holder weight. The fourth input rests on an identity. Bitcoin's holders together own all of it, so their average weight is

$$w_t \;=\; \frac{C_t}{W_t}, \tag{2}$$

Bitcoin's capitalization divided by the total wealth of its holders. This is also the quantity that drives the premium in Merton (1987): the size of the asset relative to the wealth of the investors who hold it.

Capitalization is observed every month. Holders' wealth is observed once: in the 2021Q3 survey owners had 19% of their financial assets in crypto on average, and taking that as their combined share (Section 8) gives $W = C/0.19$. For every other date W_t must be estimated, and the result depends on who counts as a holder. We use two readings. They are alternatives, and the band below takes the more demanding one as its upper edge.

Reading 1: the average owner. The holders are everyone who owns crypto. Their wealth is the survey value, scaled by the number of owners and by wealth per owner:

$$W_t \;=\; \frac{C_0}{0.19}\cdot\frac{q_t}{q_0}\cdot k_t, \qquad w_t^{\text{avg}} \;=\; \frac{C_t}{W_t + I_t/0.015}. \tag{3}$$

Here 0 denotes 2021Q3, q_t is the surveyed share of adults who own crypto, and k_t is wealth per owner relative to 2021, an assumed path from 0.25 in 2012 to 1.4 in 2026. I_t is the Bitcoin held through funds and by public companies: the Grayscale trust from 2013, corporate treasuries from 2020 and the spot ETFs from January 2024. Their investors are assumed to have 1.5% of their wealth in it (Appendix B). We compute the weight with I_t and without, which gives a band: 3% at both ends in 2013, 15–17% in 2017, 12–19% in 2021 and 8–23% in 2026 (Figure 1d).

Reading 2: the early holder. An average hides who held the coins. In 2013 Bitcoin was worth about $1 billion and the ownership rate implies some 600,000 owners. They cannot all have been fully invested: that would put each owner's entire wealth at $1,700. Most coins sat instead with a few miners and early adopters for whom Bitcoin was nearly everything they owned, and it was they who had to be willing to keep it. In

this reading the holders are that group alone. Their weight is 1 while ownership was no higher than in the 2018 survey (1.2% of adults), and it falls in proportion as ownership grows:

$$w_t^{\text{early}} \;=\; \min\!\left(1,\; \frac{q_{2018}}{q_t}\right), \tag{4}$$

with q_t as in equation (3). The weight is 0.5 in mid-2019 and 0.1 from 2021, by then below the average owner's.

Where the fully invested era ends is a judgment. We end it at the 2018 survey: regulated Bitcoin futures began trading in December 2017, and the price gaps across exchanges that mark a segmented market are documented through February 2018 (Makarov and Schoar 2020). Ending it at the 2014 survey changes one period: the upper edge for 2015–17 falls from 105% to 79%, and delivered returns exceed it by 81 points, not 55 ($t = 1.7$). Nothing since 2018 moves by more than three points, and the 2026 estimate is unaffected (Appendix D.2).

The band. Because several inputs are ranges, the required return at each date is a band, not a number. Its two edges are:

- *Lower edge*: the average owner with every input at its low end. Risk aversion is 1, the weight counts fund and treasury investors, mortality is 0.3 points, and the market term is cut by 15%.
- *Upper edge*: the larger of two numbers. (a) The average owner with every input at its high end: risk aversion 2.8, the weight of surveyed owners only, mortality 3.5 points, the market term raised by 15%. (b) The early holder's requirement, $r_f + w_t^{\text{early}}\sigma_t^2$.

Table 4 works the edges out at five dates, and panel (a) of Figure 2 shows them month by month. The early holder sets the upper edge in every month through 2016, in some months of 2017–19, and in none since. The band is an envelope, wider than a confidence interval: the inputs are unlikely to have sat at the same extreme together. Its midpoint is the average of the two edges: a convention, not an estimate.

Step 2: what Bitcoin delivered. The required return is an arithmetic average, so delivery is measured the same way: as twelve times the average monthly return. A compound return would understate it by the volatility drag, 20 to 100 points a year. A single month is too noisy to plot: annualized, months run from -470% to $+5{,}400\%$. Panel (a) of Figure 2 therefore smooths them: the black line is the average return from eighteen months before each date to eighteen months after it, and plays no part in the test. The circles are the period averages of Table 5, which the test compares.

Step 3: the test. The required return computed at a month-end is quoted per year, as an interest rate is, and applies from that date on. We convert it to a one-month return and pair it with what Bitcoin returned in the month that follows. At the end of December 2023, for example, the midpoint was 22.5% a year, or 1.9% for the month; Bitcoin returned 0.9% in January 2024, so that month's gap is -1.0 point. There are 172 such pairs, none overlapping, from January 2012 to April 2026. Table 5 averages the required and the delivered returns over the same months, for the whole sample and by three-year period, long enough to average over most of a boom and a bust (Appendix D.2), and quotes both as annual rates; Figure 3 plots the period gaps.

- *Over the whole sample delivered and required returns are consistent, but the fit is loose.* Bitcoin delivered 118% a year. The band averaged 20–111%, with midpoint 64%. The gap to the midpoint is 54 points, with a standard error of 39.

Table 4: The required-return band worked out at five dates.

	Mid-2013	Mid-2017	Mid-2019	2021Q3	May 2026
Inputs at the date					
Risk-free rate, %	2.5	2.3	2.0	1.5	4.5
Volatility, %	151	80	90	77	42
Beta	3.00	2.15	0.70	1.30	1.75
Average owner's weight, %	3	15–17	20–25	12–19	8–23
Early holder's weight, %	100	100	51	11	10
Required return, % a year					
Lower edge: average owner, low inputs	21	20	21	13	12
Upper edge, the larger of					
(a) average owner, high inputs	42	46	**64**	**42**	**27**
(b) early holder	**231**	**67**	43	8	6
Midpoint	126	43	43	28	20

Bold: the number that sets the upper edge. The market term, beta times the 4% ERP, moves 15% either way, as the 2026 range of beta (1.5–2.0) does around 1.75. The May 2026 column uses that month's inputs; the 2026 estimate of Table 3 takes the risk-free rate at 5.0% and the high weight at the surveyed 19%. The early holder's requirement is the bound of Martin (2017) for a fully invested investor with $\gamma = 1$, which includes the market and mortality terms. Formulas: Appendix D.2.

- *The early years favor the early holder.* The lower edge, the average owner, leaves 98 points unexplained and is rejected ($t = 2.5$); the upper edge, the early holder until 2018, leaves 7. The gap to the midpoint is +120 points in 2012–14 (standard error 169) and +98 in 2015–17 (46), the one period in which the midpoint is rejected.

- *Since 2018 the two agree.* The gaps to the midpoint in the three periods are +30, +3 and +10, each well inside one standard error, and the gap to the upper edge is −1.

Why did Bitcoin deliver more than the midpoint in its early years? Three readings fit, and the data cannot separate them. Concentration may have been greater than the back-cast assumes: the returns of 2012–17 are what a holder with γw of about 1.2 (standard error 0.5) would require, that is, nearly all her wealth in Bitcoin at a risk aversion near one, against back-cast weights of 6–12%. This is the early holder of equation (4), whom the returns do not reject. Early holders may have been paid for risks the equation does not price and the census cannot measure: that Bitcoin itself would fail, that exchanges would (as Mt. Gox did in 2014), and illiquidity. And the required return was falling, which by itself lifts the price: an asset whose discount rate falls delivers more than it required while the fall lasts, as US equities did when their valuations rose (Fama and French 2002).

The gap closed for two reasons. The band narrowed from over 200 points wide to about 15 as the two readings of who holds Bitcoin converged, and volatility fell, so each year of returns measures the average more sharply. A caution: a single month, November 2013, when the price rose 451%, contributes 31 of the 118 points, and without calendar 2013 the full-sample gap to the midpoint is 12 points (standard error 21).

What the test cannot show. First, it has little power: a standard error of 24 points since 2018 cannot tell a 31% requirement from a 21% or a 41% one, nor the required return from the age rule, whose gap is 48 points over the full sample and 1 since 2018. Second, it is a test of level, not of timing. Month by month the required return's correlation with the following return is zero (0.17 for the age rule), and over five-

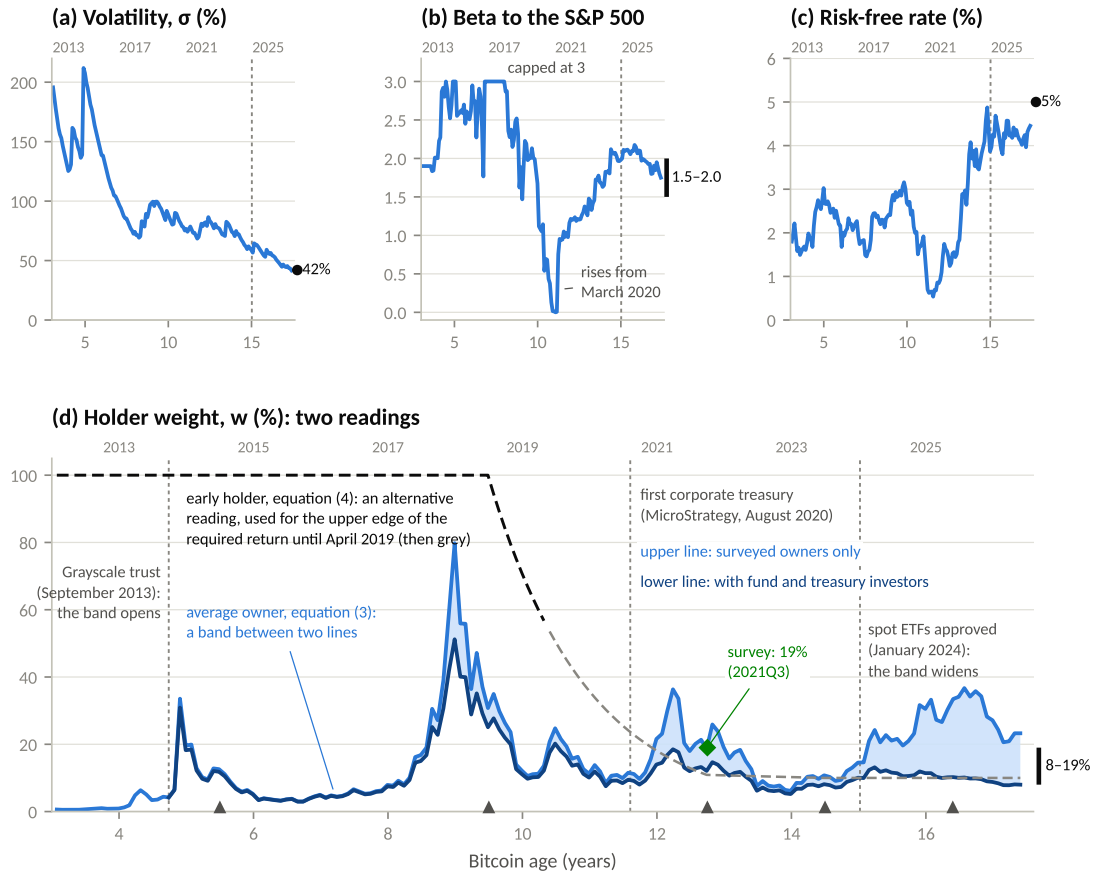


Figure 1: The four inputs of equation (1) that move with time, at each month-end. (a) Volatility and (b) beta, as defined in the text. (c) The ten-year Treasury yield. (d) Holder weight: the average owner of equation (3), a band from mid-2019, and the early holder of equation (4). Triangles: the surveyed ownership rates. Right edge: the 2026 values of Table 3. Dotted line in every panel: spot-ETF approval, January 2024.

year windows the age rule fits the path far better (Appendix D). The reason is in the construction: trailing volatility and a weight with capitalization in its numerator make the band highest at market peaks, when subsequent returns are lowest. The equation is a date-by-date restriction, not a timing signal.

What drove the requirement. Panels (b) and (c) of Figure 2 split the midpoint into the four terms of equation (1). Read against panel (a), they show that each spike in the required return is a spike in the concentration term. That term was about 87% of a requirement near 126% a year in 2013, and still about 70% in 2021. After the dilution of the holder base and the fall in volatility, the mix in April 2026 is market risk 35%, concentration 34%, the risk-free rate 22% and mortality 9%. Bitcoin's required return is now mostly the price of ordinary market risk and the risk-free rate, the profile of a high-beta risk asset, with a concentration premium that is material but no longer dominant. It is not the profile of gold: since 2020 Bitcoin's monthly returns have a correlation of 0.54 with the S&P 500 and 0.04 with gold (Appendix E). (The midpoint ends at about 20% in May 2026, not the 18% of Section 4, because it weights the band's corners equally, which the propagation of Appendix D does not.)

The age rule, on this account. Both the required return and the age rule fell from over 100% to tens of percent. The age rule is best read as a reduced form of the required return: right on average while the inputs

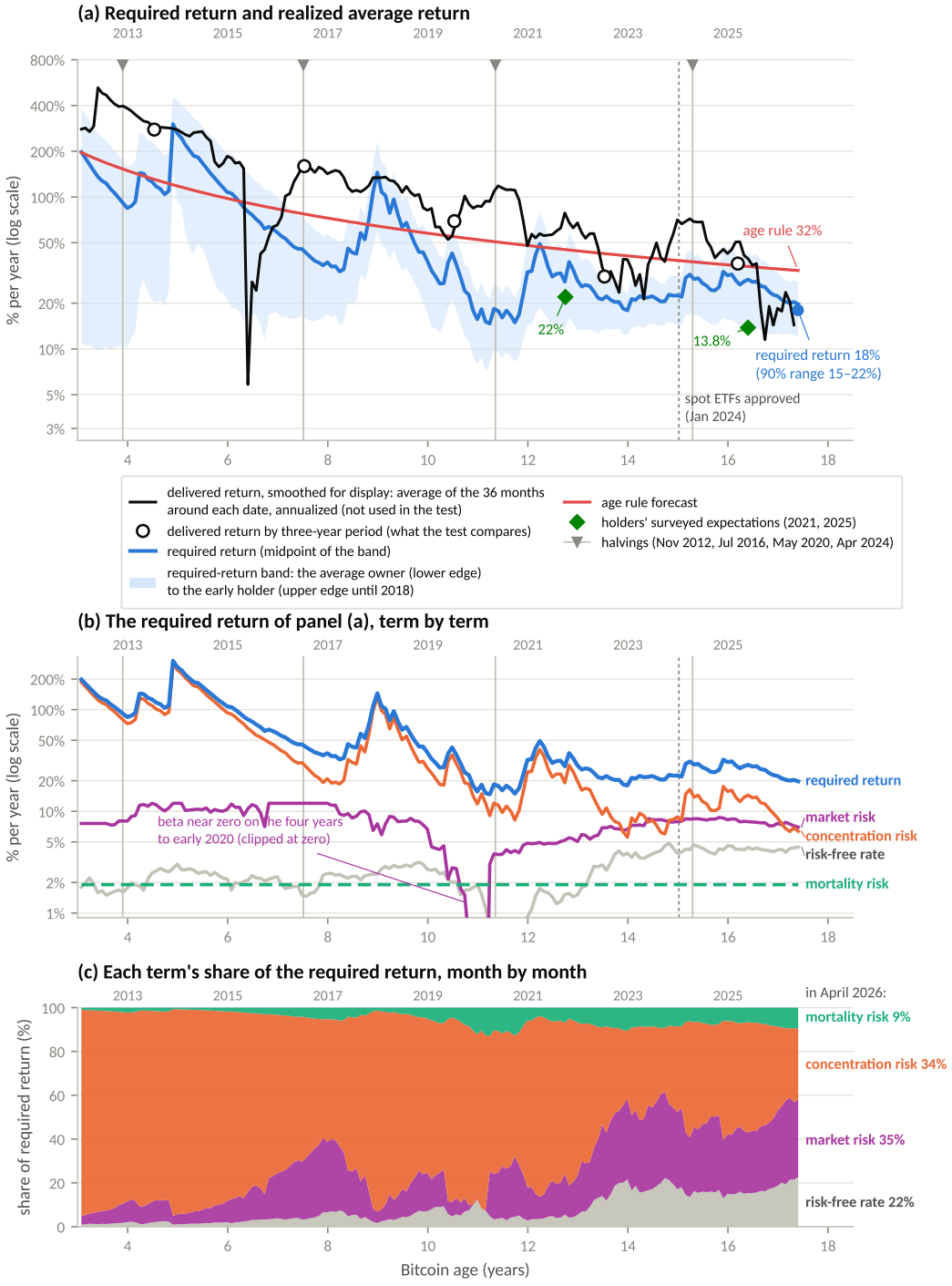


Figure 2: The required return since 2012. (a) The band and its midpoint: equation (1) at each month-end, from the inputs of Figure 1 and the edges of Table 4. Black line: Bitcoin's monthly returns, annualized and smoothed (Step 2); not used in the test. Circles: the period averages of Table 5, which the test uses. Other marks as labelled. (b) The midpoint of panel (a) split into the four terms of equation (1); mortality is held at 1.9 points and concentration is the remainder. (c) Each term's share of the midpoint, month by month; the labels give April 2026.

Table 5: The monthly test: delivered against required return, percent per year.

		Required		Delivered minus			
Period	Delivered	Band	Midpoint	Lower edge	Midpoint (s.e.)	Upper edge	Age rule
2012–26, all	118	20–111	64	+98	+54 (39)	+7	+48
2012–14	278	31–298	157	+247	+120 (169)	−21	+143
2015–17	160	20–105	62	+140	+98 (46)	+55	+82
2018–20	69	19–60	39	+50	+30 (48)	+9	+15
2021–23	30	14–39	27	+16	+3 (39)	−9	−12
2024–26	36	14–38	26	+22	+10 (33)	−2	+1
2012–17	219	25–202	110	+193	+109 (87)	+17	+112
2018–26	46	16–46	31	+30	+15 (24)	−1	+1

Each observation pairs a month-end's required return with Bitcoin's simple return over the following month; all figures are twelve times the monthly average. There are 172 months: 36 in each three-year period and 28 in the last, which ends in April 2026. Delivered minus required is the average monthly difference, with its standard error in parentheses for the midpoint. Age rule: the one-month form of the rule, with the exponent held at its 2026 value, 5.62.

happened to fall roughly as one over age, with no structure to say whether they will go on doing so. They have not: volatility and the holder weight fell with integration, not with age, and the market term rose with it. The age rule's 33% for mid-2026 sits above the top of the required band (27%), and in windows beginning after the power law's 2019 publication Bitcoin fell short of the rule by 5–8 points a year (Papadogiannis 2026a).

Premium or belief? A large holder weight has two possible sources. Few investors can or will hold the asset, so each holds a lot and the price must pay them a premium (Merton 1987). Or investors pile in because they expect a great deal, the price rises to meet the belief, and the return does not arrive. The weight alone cannot tell the two apart; the returns can. They ran above the midpoint before 2018 and have matched it since, which favors the premium reading over the sample as a whole. It does not hold at every date: the required return was highest at the 2013, 2017 and 2021 peaks and the returns that followed were below it, which is what belief-inflated weights at peaks would produce.

The cross-section. The typical altcoin, more volatile and likelier to die, required more than Bitcoin and delivered less: since 2018 the median one has lost 27% a year compounded (Section 6). Bitcoin, where delivered and required returns meet, is the winner of its cohort (at age ten the median coin born 2010–14 had compounded at 50% a year, Bitcoin at 233%), so the required return is validated on a selected path.

6. THE SAME EQUATION FOR ETHER AND OTHER CRYPTOASSETS

Nothing in equation (1) is specific to Bitcoin. For another cryptoasset three of the inputs are measured as they are for Bitcoin: volatility and the beta to equities from the asset's prices, and the mortality term from the census, by size class. The fourth, the holder weight, needs one more step.

The holder weight of another cryptoasset. For Bitcoin the holder weight w of equation (2) was anchored on surveys and carried through time (Sections 4 and 5). What the surveys report is the share of an owner's wealth in crypto as a whole, which we read as Bitcoin's weight. This section makes that reading exact and extends it to other coins. As a group, crypto owners hold every cryptoasset in proportion to its

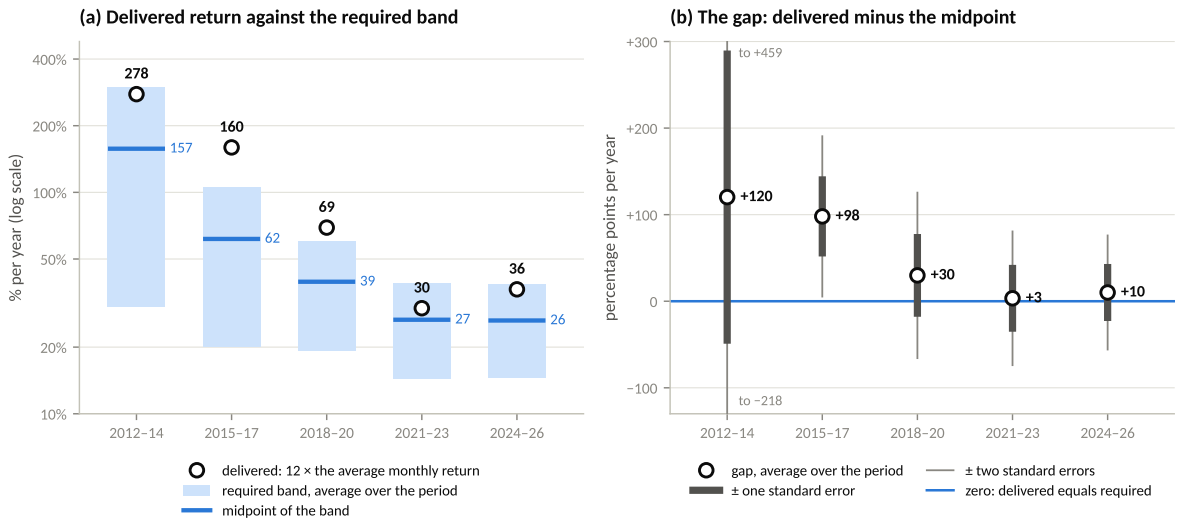


Figure 3: The monthly test by three-year period (Table 5). (a) The average delivered return (circles) against the average band and its midpoint; logarithmic scale. (b) The gap, delivered minus midpoint, with one and two standard errors.

capitalization, since each coin must be held by someone, so the average owner's crypto holding is the crypto market itself. We measure that market on the 84 cryptoassets in the Coin Metrics data that report a price and a capitalization, stablecoins excluded, each weighted by its capitalization: in April 2026 it was 73% Bitcoin, 14% Ether and 13% other coins. The concentration term of equation (1) pays the holder for the risk the asset adds to her own portfolio, which is the covariance of the asset's return with the return on what she holds. If she holds one asset, at weight w, that covariance is $w\sigma^2$, the form in equation (1). If she holds the crypto market, at the same weight w, it is w times the asset's covariance with that market. Equation (1) therefore applies to asset i with the weight

$$w_i = \kappa_i w, \qquad \kappa_i = \frac{\mathrm{Cov}(r_i, r_c)}{\sigma_i^2} = \rho_i \frac{\sigma_c}{\sigma_i}, \tag{5}$$

where w is the weight of equation (2), r_i and r_c are the returns of the asset and of the crypto market, σ_i and σ_c their volatilities, and ρ_i their correlation. In words, when the asset moves by 10%, the owner's crypto holding moves by κ_i times 10% on average, so an owner with a share w of her wealth in the crypto market bears the asset's risk as if she held $\kappa_i w$ of her wealth in that asset alone.

The values of κ. We estimate κ on monthly returns over the four years to April 2026. For Bitcoin it is 1.0 (1.04, and 0.99 since 2018): Bitcoin is three-quarters of the market and the rest moves with it, so the market moves one for one with Bitcoin, and $w_i = w$: the weights of Sections 4 and 5 stand as they are. For Ether κ is 0.7. Ether's correlation with the market is 0.91, but over those four years its volatility was 74% against the market's 57%, so the market moves by 7% when Ether moves by 10%. For the median large altcoin κ is 0.47 (median correlation 0.74, median volatility 86%). With the 2026 weight of Table 3, w of 8–19%, the holder weights are therefore 8–19% for Bitcoin, 6–13% for Ether and 4–9% for a large altcoin. Their concentration terms are still larger than Bitcoin's, because their variances are higher.

Worked example: Ether. At the midpoint of each range w is 13.5%, so Ether's weight is $0.7 \times 0.135 =$

9.5%. With a risk aversion of 1.9 and a volatility of 64%, the concentration term is $1.9 \times 0.64^2 \times 0.095 = 7.4$ points. The market term is Ether's beta of 2.35 times the ERP of 4%, or 9.4 points, and the mortality term is Bitcoin's, 1.9 points, since both are top-twenty assets. With the risk-free rate of 5.0% the sum is 23.7%. Drawing the inputs over their ranges, as for Bitcoin, gives a median of 23% (90% range 19–29%; envelope 16–35%). Table 6 sets out the same sum for each asset.

One date, not a path. Table 6 is for 2026 only: the weight is the 2026 value, volatilities are those of May 2026, and betas and κ are estimated over the four years to April 2026. The path of w built in Section 5 belongs to the owner, not to the coin, so it would serve every asset; carrying another asset back through time would need only its own paths of κ, volatility and beta. We have not done this, because the record could not test such a path: since 2018 Ether has delivered an average of 48% a year with a standard error of 32. Its requirement was higher in earlier years. Ether's volatility has fallen as Bitcoin's has, from 172% at the end of 2017 to 99% in 2021Q3 and 64% in May 2026 (Bitcoin: 98%, 77% and 42%).

The floor. The risk-free rate, the market term and the mortality term need no holder weight. Together they are what an investor requires who holds the asset at a negligible weight: 14% a year for Bitcoin, 16% for Ether, 15% for a large altcoin and 30% for a small coin. Small coins have no price history in the data, so no κ can be estimated and the floor is all we report for them. Their death risk alone is 16–17 points: the smallest quartile of the census dies at 17.4% a year, with almost nothing recovered.

Table 6: The required return of four cryptoassets in 2026, built term by term from equation (1). Percent per year; required returns are arithmetic averages.

	Bitcoin	Ether	Large altcoin	Small coin
Inputs				
Volatility, σ	42	64	76 (66–88)	100–130, assumed
Beta to the S&P 500, β	1.5–2.0	2.0–2.7	1.5–2.7	as large altcoin
κ, equation (5)	1.0	0.7	0.47 (0.41–0.56)	not measured
Holder weight, $w_i = \kappa_i \times w$, with w of 8–19%	8–19	6–13	4–9	not measured
Required return, term by term				
Risk-free rate	5.0	5.0	5.0	5.0
+ Market risk, $\beta \times$ ERP	7.0	9.4	8.4	8.4
+ Mortality risk	1.9	1.9	2.0	16.6
= Floor: a holder with a negligible weight	14 (12–16)	16 (14–18)	15 (13–18)	30 (28–32)
+ Concentration risk, $\gamma\sigma_i^2 w_i$	4.5	7.4	7.0	not measured
= **Required return, average crypto owner**	**18** (15–22)	**23** (19–29)	**22** (18–29)	not measured
Delivered, 2018–26				
Average (standard error)	46 (23)	48 (32)	37 (33)	not observable
Compound	28	8	−2	not observable

The four terms are computed at the midpoint of each input's range: w of 13.5% (range 8–19%, Table 3), risk aversion 1.9 (1–2.8), ERP 4%, and a mortality term of 0.3–3.5 points for Bitcoin and Ether, 1–3 for a large altcoin (hazard 3.2% a year) and 16–17 for a small coin (17.4% a year) (Section 3). The floor and the required return are medians over the input ranges, with 90% ranges in parentheses (Appendix D.1); a median can lie up to 0.3 points below the sum of the terms above it. For inputs, parentheses give interquartile ranges. Large altcoin: the 57 surviving altcoins with four years of prices; inputs are medians over the last four years, volatility as for Bitcoin. Delivered: the monthly measure of Table 5 over the 100 months from January 2018, when required returns were higher than in 2026, so the two panels are not a test; for altcoins an equal-weighted basket of 84 altcoins, rebalanced monthly and including the 25 that died.

What altcoins delivered. The census keeps the coins that died, so the record carries no survivorship bias. An equal-weighted basket of the 84 altcoins with a price history, 25 of which died, earned an average

of 37% a year over 2018–26, with a standard error of 33. The basket, rebalanced monthly, compounded at -2% a year (-3% if a coin whose price feed stopped is counted as a total loss). Coin by coin the result is worse: the median altcoin lost 27% a year compounded, 77% of them lost money, and 7% did better than Bitcoin. A high average with a negative compound return is the signature of a lottery: a few large winners carry the mean while the typical holder loses, the pattern documented for lottery-like stocks (Barberis and Huang 2008).

What this section can and cannot say. The holder weight is that of the average crypto owner, who holds the whole market; an owner concentrated in one coin requires more, and no survey measures such holders. For small coins the paper offers the floor and one fact: half of the smallest quartile is dead within 3.3 years of listing (Figure 7). Down the size ladder the required return rises, because death risk and volatility rise, while the buy-and-hold record worsens. Bitcoin is the one cryptoasset for which delivered and required returns have met, which is a statement about its holder base and its survival, not about the equation.

7. FOR THE ALLOCATOR

From required to expected. The sections so far are about what holders require. An allocator needs a return to expect. The two coincide under one hypothesis, which this section adopts: *Bitcoin is priced in equilibrium, at the return its holders require* (Section 2). Section 5 gives it some standing: fourteen years of returns are consistent with the required return, and the market is more integrated now than at any earlier date. The 18% (9.5% compounded) is the required return on today's inputs and moves with them: a further fall in the requirement would lower later returns but lift the price while it lasts. J.P. Morgan Asset Management (2026b) assumes 10.0% compounded and 20.9% on average, on a volatility of 55% (Appendix E.3).

The hurdle. A diversified allocator should require of Bitcoin what its contribution to her portfolio's risk costs her. She holds it at a small weight, so the concentration term of equation (1) drops out:

$$\mu_0 \;=\; r_f + \beta\,\mathrm{ERP} + \lambda(1-\delta). \tag{6}$$

The first two terms are **11–13% today**. Of the mortality term, 0.3–3.5 points, she should charge the part she judges systematic rather than diversifiable. The 2022 clustering of elite deaths (Terra, Celsius, FTX within six months) argues it is partly systematic; we would charge 1–2 points, which makes the hurdle 12–15%. It is not a forecast but the bar Bitcoin must clear to belong in her portfolio at more than market weight.

Is it worth holding at all? For the smaller cryptoassets the buy-and-hold record says no: even a diversified holder requires 15–30% a year, and the median altcoin has lost 27% a year since 2018 (Section 6). For Bitcoin's concentrated holders the 18% is what they require and, in equilibrium, what they get, so it carries no alpha for them. A diversified investor does not bear the concentration risk they are paid for, and at a market weight under 1% she barely bears the death risk either: a jump to zero would cost her less than 1% of wealth. Her hurdle is 12–15%, and the asset offers about 18%. The difference of 3–6 points is a rent: the premium a still-concentrated holder base demands, open to an investor who holds the asset as a small part of a diversified portfolio (Merton 1987). The rent is what remains of the concentration term, and it shrinks as holders become less concentrated. It is the least certain part of the estimate: zero at the bottom of the 90% range, and absent if holders' own 2025 expectation of 14% is the right one.

The sizing rule. Equation (1), inverted, sizes the position. An investor who expects Bitcoin to return $\hat{\mu}$

and has risk aversion γ should hold

$$w^* \;=\; \frac{\hat{\mu}-\mu_0}{\gamma\sigma^2}, \tag{7}$$

the return she expects above her hurdle, divided by what each unit of weight costs her in risk. Worked example: an investor with $\gamma = 3$ who takes the 18% required return as her expected return, at today's $\sigma = 42\%$ ($\sigma^2 \approx 0.18$), a beta of 1.75 and the low reading of mortality: $w^* = (18-5-7-0.3)/(3\times 0.18) \approx$ **11% of financial wealth**. An investor who believes the age rule's 32% gets $w^* \approx 37\%$.

The Black–Litterman translation. Equation (7) is the Black–Litterman rule for a single view (Black and Litterman 1992; He and Litterman 1999). That rule starts from the return implied by holding the asset at its market weight w_m, here μ_0. A view $\hat{\mu}$ held with confidence θ, between 0 and 1, moves the expected return part of the way to the view, and the weight in proportion:

$$\bar{\mu} \;=\; \mu_0 + \theta\,(\hat{\mu}-\mu_0), \qquad w - w_m \;\approx\; \frac{\bar{\mu}-\mu_0}{\gamma\sigma^2} \;=\; \theta\,\frac{\hat{\mu}-\mu_0}{\gamma\sigma^2}. \tag{8}$$

This paper supplies both ends of the gap for Bitcoin: μ_0, the 12–15% a diversified investor requires at market weight, under 1% of global financial wealth, and $\hat{\mu}$, the 18% its concentrated holders make it offer. Full confidence gives the 11% of the example; half confidence gives about 5%.

Buy and hold, or rebalance. Equations (1) and (7) are written on the arithmetic return, and who collects that return depends on what the investor does. A position bought and never rebalanced grows at Bitcoin's own compound rate, so its typical outcome is about 9.5% a year; the 18% is the average across paths, lifted by a few very large ones. On that footing Bitcoin offers little more than a global equity index (about 8% compounded) at three times the volatility. A position rebalanced to a fixed weight is different: the slice contributes its arithmetic return, and the volatility drag is paid on the portfolio, not on Bitcoin (Booth and Fama 1992). Five percent in Bitcoin, held against the S&P 500 and rebalanced, raises the portfolio's expected return by about 0.45 points a year and its volatility from 14.1% to 14.6%, which costs about 0.07 points in extra drag. The premium of this paper is therefore a rebalancer's premium; the buy-and-hold investor should plan on the compound figure. The drag (Table 7) is half the variance (Jacquier, Kane, and Marcus 2003), so it grows with the square of volatility: it takes one to two points from equities and gold, half of Bitcoin's required return, and nearly nine-tenths of Ether's. On average returns Bitcoin's Sharpe ratio is the highest in the table; on compound returns it is 0.11, below every equity index (0.15–0.21), and Ether's is negative.

Against equities, standing alone. The S&P 500's expected excess return is the ERP, 4% (6% in the high case). Its volatility was 14% a year over January 2012 to April 2026 (Table 7) and is 16–20% over the long history (Dimson, Marsh, and Staunton 2002), so its Sharpe ratio (Sharpe 1994) is 0.20–0.29, and 0.30–0.43 in the high case. Bitcoin's is $(18.3-5)/42 \approx 0.32$, and about 0.27 once the mortality term, which is compensation for a loss and not a return, is taken out. If the required return is paid, Bitcoin offers about the same reward per unit of risk as equities on their recent volatility, and somewhat more on their long-run volatility; where there is an excess, it is the concentration premium.

Which equity index. The choice changes little. On the same pricing a global index, the Nasdaq-100 and the Russell 2000 all have a required compound return of about 8%, because for the higher-beta indices the extra premium and the extra drag nearly cancel (Table 7). Realized returns over those months were much further apart: about 20% a year for the Nasdaq-100, 15% for the S&P 500 and 11–12% for the global index

Table 7: Average and compound required returns in 2026.

		Required return and risk, % a year				Sharpe ratio	
	Model	Average	Volatility	Drag	Compound	Average	Compound
Global equities (MSCI ACWI)	CAPM	8.8	13.8	1.0	7.8	0.28	0.21
S&P 500	CAPM	9.0	14.1	1.0	8.0	0.28	0.21
Nasdaq-100	CAPM	9.5	17.2	1.5	8.0	0.26	0.18
Russell 2000	CAPM	9.7	19.1	1.8	7.9	0.25	0.15
Gold	MCM	5.5	15.5	1.2	4.3	0.03	−0.05
Bitcoin	MCM	18.3	42	8.8	9.5	0.32	0.11
Ether	MCM	23.4	64	20.5	2.9	0.29	−0.03

Required returns in nominal US dollars, not histories (in euros: Appendix E.4). Model: the equity indices are priced by the CAPM, the 5% risk-free rate plus the 4% ERP times each index's beta on the S&P 500 (0.95, 1, 1.12 and 1.18), with no separate size or style premium. Gold, Bitcoin and Ether are priced by the MCM equation (1) (Table 6; for gold Appendix E, where the figure is an upper bound). Volatility: of monthly returns in US dollars. For the equity indices and gold, the whole of January 2012 to April 2026 (ETF total returns); for Bitcoin and Ether, whose volatility has been falling, recent months weighted more (Appendix A.7). Drag: half the variance. Compound: average less drag, the growth rate of a position never rebalanced. Sharpe ratio: the average or compound return, less the 5% risk-free rate, divided by volatility. The ratio on compound returns is descriptive; Sharpe's (1994) measure is defined on average returns.

and the Russell 2000. We do not carry those gaps forward, for the reason we do not carry Bitcoin's history forward: a realized return is not a required one, and indices with similar risk cannot be required to pay very different premiums. Nor is the US lead a regularity: US equities trailed the developed markets outside North America in three of the five decades since 1970, and of their lead of 4.6 points a year since 1990 all but a statistically insignificant 1.2 points came from a rise in relative valuation (Asness, Ilmanen, and Villalon 2023). These are US-dollar returns; in euros the global index was less volatile than the S&P 500 (12.6% against 13.8%; Appendix A.7).

What has changed is the diversification argument. Bitcoin's correlation with equities, about 0.2 before 2020, has been about 0.5 since (Appendix A; Copestake, Furceri, and Terracciano 2024), so about two-fifths of its required return is now payment for a risk the equity investor already holds. What this paper removes is not the case for a small Bitcoin position but the case built on 32% and on zero correlation.

In Sharpe units. Dividing equation (1) by σ turns the required return into a required reward per unit of risk, and because $\beta = \rho\,\sigma/\sigma_m$ the market term simplifies:

$$\frac{\mu - r_f}{\sigma} \;=\; \rho\,\mathrm{SR}_m \;+\; \gamma\sigma\,w \;+\; \frac{\lambda(1-\delta)}{\sigma}, \tag{9}$$

where ρ is the asset's correlation with the market and SR_m the market's Sharpe ratio. For a fully diversified holder of an asset that cannot die the right side collapses to $\rho\,\mathrm{SR}_m$, the CAPM in Sharpe units: no asset can offer more reward per unit of risk than the market times its correlation with it. The concentration term, $\gamma\sigma w$, is what a holder adds when she owns a lot; σw is the position's contribution to her portfolio volatility, so the ratio depends on the size of the position. Today the market term is about 0.15 (7 points over 42%; a correlation of about 0.5 at a market Sharpe ratio near 0.25–0.3), the concentration term about 0.11 ($1.9 \times 0.42 \times 0.135$) and mortality about 0.05 (1.9 points over 42%): a required Sharpe ratio of about 0.31. Once holders are diluted toward market weight ($w \to 0$) the requirement falls to $\rho\,\mathrm{SR}_m$ plus the mortality term, about 0.20 at today's volatility. Bitcoin's Sharpe ratio exceeds that only because its holders

are still concentrated, and integration should push it down. For the allocator, an asset improves a diversified portfolio's Sharpe ratio if and only if its own exceeds ρ times the portfolio's, here about 0.32 against about 0.13, so a small position adds today.

What about the halving cycle? The four-year cycle does not enter the required return. The halving changes supply, which is not one of the risks a holder is paid for, and its schedule is known years ahead, so in equilibrium it is in the price and cannot carry a predictable excess return. The past cycles were real (Figure 2 marks the halvings). In this paper's terms they are realized returns swinging around the required return, which average out, as they have since 2018.

What about the upside stories? The most common is monetary debasement. Returns in this paper are nominal, and the 5.0% risk-free rate of equation (1) already contains the inflation that bond investors expect; only inflation beyond that would be news. Tokenization, payments by AI agents and a flight from government debt under fiscal dominance are claims about future demand, and the equation prices risk. A story that investors already believe is in today's price and does not raise the return from here. If the market underestimates one and it arrives, Bitcoin will return more than required, once. Afterward its required return is lower, not higher: less concentrated ownership shrinks the concentration term, and an asset valued as a hedge is one that investors accept a lower return to hold, as with gold (Appendix E).

Appendix E applies the same condition to gold, where it reproduces a known result (a required return near the risk-free rate; Barro and Misra 2016), and spells out what "digital gold" would mean in return terms: about 14% a year at gold's volatility and holder dilution, because Bitcoin would still carry its beta and its death risk, and gold's 5–6% only if it shed both.

8. LIMITATIONS AND THE MEASUREMENT AGENDA

1. **The holder weight before 2021 is bounded, not measured.** The portfolio-share surveys begin in 2021; everything earlier rests on a market-clearing back-cast for the average owner and on an assumed fully invested early holder, and the 2026 lower bound on an assumed weight for fund and treasury investors (Appendix B). The most valuable upgrade would be wealth-weighted portfolio shares from each survey wave.

2. **Coverage-onset selection in the graveyard.** Assets enter the panel only after becoming significant (Section 3). This biases the data *toward* Lindy, which makes the rejection conservative, and leaves the elite estimates untouched; but population hazard *levels* at ages under two are floors.

3. **Death timing and classification.** Feed and volume deaths lag economic death by months; four elite "deaths" are contested zombies. All classifications, dates, recovery fractions, and confidence flags are in Appendix C, and the headline elite hazard moves only from 3.5% to 3.15% when the contested cases are dropped.

4. **Bitcoin's own existential era is unmeasurable.** No cross-section existed in 2010–2014, when Bitcoin had no peers, to price its risk of failure. The band's upper edge in those years is the requirement of a fully invested holder, which includes mortality (Section 5); the 2026 estimate does not rest on them.

5. **Survey-to-wealth pairing.** The participation series is US-based and the wealth aggregates global; re-running the clearing calculations with US household wealth against US participation moves the bands by amounts well inside their width (Appendix D). The survey's 19% is an average of household shares, not the combined share of equation (2); a combined share of 12% would scale the average owner's weight down in proportion and put the central estimate for 2026 at 17.2%; ownership anchored at 3% in 2021 instead of 11% would put it at 15.7% (Appendix D). The surveys also measure crypto as a whole, which we read as Bitcoin's weight (Section 4); a back-cast on total crypto capitalization leaves the test unchanged and lowers the 2026 estimate by about half a point (Appendix D).

6. **Risk aversion is not measured for holders.** The upper end of the range for γ comes from studies of ordinary households; the lower end, 1, is our reading of surveys that find crypto owners more tolerant of risk (Hayashi and Routh 2025; Stix 2021).

7. **The holder weight is endogenous to beliefs.** A weight chosen on over-optimistic expectations satisfies the same first-order condition as one forced by segmentation, so the concentration term recovers the holder's subjective hurdle; whether that hurdle was a fair premium is adjudicated by realized returns (Section 5) and by the option-implied bound, not by the equation. The 2026 estimate reads the ETF-era weights as participation-driven, a reading that only later returns can check.

8. **The equation is complete only for the holder it describes.** The premia and yields it leaves out (Section 2) are not priced separately; option prices, stated expectations and the monthly test bound their net effect only loosely, and a few points in either direction cannot be excluded.

9. **The validation is on a survivor, and has limited power.** The monthly test does not reject the required return, and since 2018 the two agree within 15 points a year, but the standard error of that gap is 24 points (about 40 over the full sample). Early returns ran above the midpoint, Bitcoin is the winner of its cohort, the band is wide, the midpoint is a convention rather than an estimate, and the required return does not track the timing of returns (Section 5). The 2025 survey answer is below the model (Section 4).

10. **The market beta is unstable.** The rolling four-year beta has ranged from about zero to 3 and the correlation from zero to 0.55 (Appendix A); the 2026 estimate uses 1.5–2.0, and a return to pre-2020 correlations would lower the required return by several points.

11. **The other cryptoassets are measured less well than Bitcoin.** In Section 6 the weights of holders in individual coins are not observed, small coins have no price history, and the delivered averages have standard errors above 30 points. Gold's holder weight and mortality allowance in Appendix E are judgments. The Bitcoin estimate depends on none of these.

9. CONCLUSION

The riddle was that Bitcoin seemed to carry three expected returns at once: 32% from its price trend, 11–16% from the CAPM, 14–22% from its holders' mouths. Only the first was ever an expected return; the second is the hurdle of a diversified investor, and the third is what holders say they expect. What was missing

is what those holders should require given the risks they actually bear, and this paper supplies it with the MCM equation: Merton's (1987) limited-participation pricing, with a market term from a rolling beta, a concentration term from surveyed portfolios, and a mortality term from a census of 1,815 cryptoassets. The graveyard shows mortality was never the main story (hazard rises with age; the top-20 jump hazard is 0.3% a year). Concentration was: it supplied most of a required return that ran above 100% a year in Bitcoin's early life. Tested month by month, that required return is in line with what Bitcoin delivered: early returns ran toward the top of its band, and since 2018 the gap is 15 points a year, under one standard error. Bitcoin's history is consistent with a very large risk premium that was paid, though the record of one survivor cannot rule out the alternative, beliefs that happened to come true (Section 5).

What remains, if the asset is priced in equilibrium, is a required return built from observable things: 40-some percent volatility, a beta to equities near 1.75, a holder base diluting toward market weight, a small and datable mortality risk, and a risk-free rate. Those ingredients say about 18% a year on average today, with a 90% range of 15–22%, or about 9.5% compounded, above the 14% that holders themselves reported expecting in 2025; the same equation, fed gold's inputs (Appendix E), returns the risk-free rate plus half a point. The age rule's 32% sits above the band, and the framework says why: it extrapolates a decline that was driven by integration, not by age, and integration has now raised one term while lowering two others.

For practitioners. First, where Bitcoin is heading. As its holders diversify, equation (1) collapses to the CAPM, and the CAPM's advice is the familiar one (Tobin 1958; Sharpe 1964): hold the market portfolio and the risk-free asset, in proportions set by one's taste for risk. In that world Bitcoin is an ordinary high-beta asset that belongs in a portfolio at its market weight, under 1%, with a required return of 12–15%. Second, where it is today. Its holders are still concentrated, so it offers about 18% a year (15–22%), 3–6 points more than a diversified investor needs. An investor willing to hold more than market weight, and to rebalance, can collect that premium, up to about 11% of financial wealth at a risk aversion of 3; it is a rent that shrinks as less concentrated holders come in. Third, what a position that is bought and never rebalanced should expect: about 9.5% a year compounded, little more than a global equity index offers, at three times the volatility, and not the 32% that its price trend extrapolates to. Throughout, do not treat Bitcoin as uncorrelated (its beta to equities is about 1.75), and charge for mortality explicitly.

Beyond Bitcoin. The MCM equation is not specific to Bitcoin: we offer it as a template for any new asset that is volatile, held by few, and able to fail. Applying it takes four steps: measure the asset's volatility and its beta to the investor's other wealth; find out who holds it and what share of their wealth they keep in it; count how often assets of its kind fail and what holders recover when they do; and add the three resulting terms to the risk-free rate. The applications to Ether, smaller cryptoassets and gold rest partly on assumed inputs, so the template is validated here on one asset only; tokenized assets, private credit and early-stage equity are natural next tests.

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DECLARATION OF GENERATIVE AI USE

Generative AI (Anthropic's Claude) was used under the author's direction to write and test the code in the replication package, to produce the figures from that code, and to draft and edit text. The conception of the research, the research design, the choice of the model, the choice of every assumption in it, the choice of figures and tables, the curation of the death classifications, the assessment of prior work and of the paper's contribution, all interpretations and conclusions, and the final text are the author's, who has verified the outputs and takes full responsibility for the content. Generative AI was not used to generate or alter data.

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Data and code availability: the full replication package (panel-construction and survival scripts, the curated asset-level panel, all result tables, and every exhibit) is maintained at `https://github.com/apapadogiannis/bitcoin-required-return`. The pipeline ends with an audit script that recomputes and checks every quantitative claim in this paper. Underlying market data are the public Coin Metrics community dataset (snapshot ending May 24, 2026), which the package downloads from its source rather than redistributing.

A. MARKET RISK AND THE RISK-FREE RATE

The market term of equation (1) is Bitcoin's beta to the equity market times the ERP, and it is added to the risk-free rate. The paper uses these three inputs in two places, the 2026 estimate and the path since 2012, and treats them differently in each.

A.1 How beta is estimated. Beta is the slope of a regression of Bitcoin's monthly log return on that of the S&P 500, both measured month-end to month-end. The index is represented by the total return of the Vanguard S&P 500 ETF (VOO, adjusted close), whose month-end ratio to the index is constant to the third decimal over the sample. The S&P 500 is used as a proxy for the rest of the holder's wealth, not as an index Bitcoin belongs to: what equation (1) prices is Bitcoin's covariance with that wealth, and the true market portfolio of all assets is not observable (Roll 1977). Three reasons support the choice. The holders whose portfolios are surveyed are US households. The ERP of A.4 is measured on the same index, so beta and premium refer to one market. And a global index gives nearly the same answer: against the iShares MSCI ACWI ETF, whose monthly returns have a correlation of 0.97 with the S&P 500, Bitcoin's beta is 1.6 on the last four years, 1.7 on the last five and 1.9 since 2020, all inside the range used.

A.2 The beta used for 2026. The 2026 estimate uses a range, 1.5–2.0, because the answer depends on how far back one looks and no window is clearly right (Table 8). On the last four years beta is 1.75; on the last five, the common convention for monthly data, 1.8; since 2020, 1.9; since 2016, 1.5; since 2012, 1.8. The range covers these. The two shortest windows give less, 1.3 and 1.4, and they are also the least precise, with standard errors of 0.5 to 0.6; we leave them outside the range and report what they would change. Every estimate is within one standard error of every other, and the standard error of any one of them, about 0.4, is larger than the half-width of the range. The range is therefore a range of point estimates, not a confidence interval. Appendix D.1 gives two sensitivities: a range of 1.3–2.2, which covers every window, and a beta drawn with its sampling error. Neither moves the central estimate by more than 0.1 point.

A.3 The beta used for the path. Section 5 needs a beta at every date since 2012, so there it is time-varying: at each month-end it is re-estimated on the previous four years (on all available data before four years exist). Panel (b) of Figure 1 shows it. Before 2020 the four-year correlation was 0.0–0.3 and the beta

Table 8: Bitcoin's beta on the S&P 500, by estimation window.

Window	Months	Beta	Standard error	Correlation
Last 2 years	24	1.26	0.64	0.39
Last 3 years	36	1.38	0.54	0.40
Last 4 years	48	1.75	0.43	0.52
Last 5 years	60	1.81	0.38	0.53
Since 2020	77	1.93	0.35	0.54
Since 2016	125	1.51	0.38	0.34
Since 2012	173	1.77	0.45	0.29

Monthly log returns, month-end to month-end, to May 2026; ordinary least squares. The index is the total return of the Vanguard S&P 500 ETF. Script `34_beta_windows.py`.

was unstable, at or above 3 in 2013–2016 only because Bitcoin was so volatile (we cap it at 3), and about zero over the four years to early 2020. Since mid-2022 the correlation has been 0.4–0.6 and the beta 1.6–2.2.

When the co-movement appeared. It did not begin with the spot ETFs of January 2024. On monthly returns Bitcoin's correlation with the S&P 500 was 0.16 over 2012–19 (0.00 over 2016–19), 0.56 over 2020–23 and 0.49 from 2024 to May 2026. The beta was 2.0 (standard error 0.4) over 2020–23 and 1.8 (0.6) since. The break is in March 2020: the correlation over the previous twelve months was -0.4 at the end of 2019 and $+0.7$ at the end of 2020. A four-year window takes four years to absorb such a break, which is why the rolling beta climbs gradually from 2020 to 2023 and is flat after the ETFs.

Why. We read the cause as the holder, not the vehicle. Beta measures how Bitcoin moves with the rest of its holders' wealth (A.1). While holders were few and nearly fully invested, there was little other wealth for it to move with. Between the 2018 and 2021 surveys the share of adults owning crypto rose from 1.2% to 11% (Appendix B), and the new holders were households and funds that also own equities. Investors who hold both sell both when their wealth or their appetite for risk falls, so the two prices move together. Three studies point the same way. From March 2020 the net purchases of stocks and of crypto by the same retail investors became positively correlated, and the stocks those investors favor co-move most with Bitcoin (Somoza and Didisheim 2022). Changes in the correlation between crypto and equities follow the participation of institutional investors, and US monetary policy moves crypto prices only when that participation is high (Copestake, Furceri, and Terracciano 2024). Spillovers between Bitcoin and equity markets rose with the pandemic (Iyer 2022). Nor does the co-movement run through use of the network: since 2020 the monthly growth of active addresses has a correlation of 0.18 with equity returns, and adding it to a regression of Bitcoin's return on equity and gold returns leaves the equity coefficient at 1.9 (script `33_activity_beta.py`). The spot ETFs continued this broadening of the holder base; they did not start it. Gold shows that a fund wrapper does not create co-movement by itself. Gold ETFs have traded in the United States since 2004, yet gold's beta on the S&P 500 is 0.10 (Appendix E): 62% of the gold stock is jewelry and central-bank reserves, and investors buy gold as a refuge when equities fall (Baur and Lucey 2010). The same change, a broader holder base, therefore lowers Bitcoin's concentration term and raises its market term (Section 2).

A test on holdings. Figure 4 sets the correlation against the share of the Bitcoin supply held through funds and by public companies, the holders most likely to own equities as well. The share was about 1% from 2014 to 2019, rose from 1.5% to 3.8% during 2020, stayed between 3.8% and 4.6% to the end of 2023,

and reached 12% in May 2026. The correlation was 0.09 over 2014–19, 0.56 over 2020–23 and 0.49 since. The first rise in these holdings coincides with the rise in the correlation. The second and larger rise, after the spot ETFs, came with no further rise. The evidence is of a step and not of a proportion: once holders who also own equities were present in size, Bitcoin moved with equities, and adding more of them has not made it move more. Two cautions apply. March 2020 was a shock to every risky asset, so the timing alone does not identify the cause, although the correlation over 2020–23 is 0.52 without that month. And the series counts disclosed holdings only: retail investors who hold both assets directly (Somoza and Didisheim 2022) are not in it.

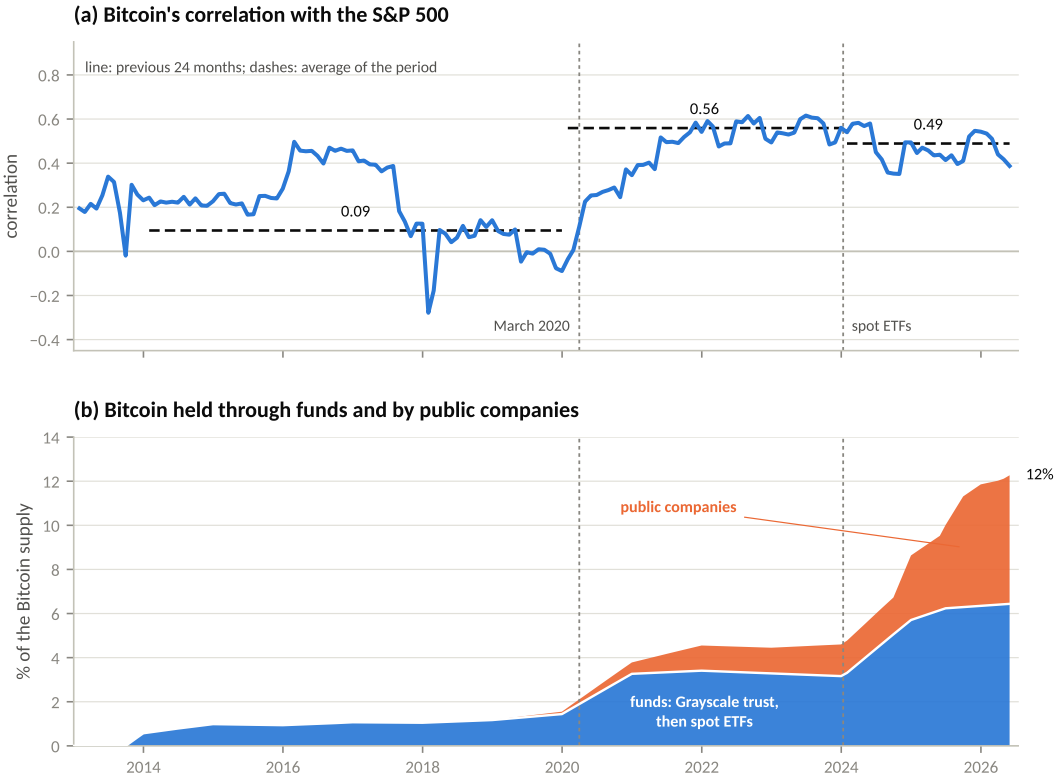


Figure 4: Bitcoin's correlation with the S&P 500 and the Bitcoin held through funds and by public companies. (a) Correlation of monthly log returns over the previous 24 months, with the average of each period. (b) Share of the Bitcoin supply held by the Grayscale trust and, from January 2024, the US spot ETFs (funds), and by public companies. Dotted lines: March 2020 and spot-ETF approval. Sources: Appendix B.3 and script `44_holders_and_comovement.py`.

A.4 The equity risk premium (ERP). The ERP is held at 4% at every date, with 6% as a high case. It is a forward-looking assumption, not an estimate that moves with the market. The premium implied by the level of the S&P 500 and its expected cash flows was 4.2% over the Treasury bond rate at the start of 2026 and 4.2–5.9% at each year-end since 2020 (Damodaran 2026); the high case of 6% is that upper end, reached at the end of 2022. Other forward-looking estimates are lower: Fama and French (2002) put the expected US premium at 2.6–4.3% from fundamentals, the J.P. Morgan Asset Management (2026a) assumptions imply about 3 points for US large caps over the ten-year yield, and realized premia have averaged more (Dimson, Marsh, and Staunton 2002). Each point of premium moves Bitcoin's required return by about 1.75 points. A reader who prefers the historical US premium of 7–8 points over bills should raise both sides of the

comparison in Section 7: equities' Sharpe ratio rises to about 0.4–0.5, and Bitcoin's market term doubles with the premium, taking its required return to 24–25% and its Sharpe ratio to 0.44–0.48.

A.5 The risk-free rate. The risk-free rate is the ten-year US Treasury yield, the usual choice for a multi-year horizon. For the 2026 estimate it is 5.0%: the yield stood at 5.3% at the end of September 2026, 4.9% at the end of October and 5.0% on November 11. For the path of Section 5 it is the month-end yield of each date (the CBOE ten-year yield index, from Yahoo Finance): 1.7% in mid-2012, 0.7% in mid-2020, 3.8% in mid-2023 and 4.5% in May 2026, the end of the sample.

A.6 Three approximations. First, for a holder with weight w in Bitcoin and the rest in the market, the exact market term is $(1-w)\beta$ times her price of market variance; at w of 8–19% the factor $(1-w)$ lowers the term by about a point, which the stated beta range absorbs. Second, a holder with risk aversion γ prices market variance at $\gamma\sigma_m^2$, which at $\gamma = 1$–2.8 and $\sigma_m = 16\%$ is 2.6–7.2%, a range that contains the 4% and 6% premia used; the market term and the concentration term are therefore priced consistently. Third, the stock index itself contains firms whose value moves with Bitcoin, so a part of the measured beta is Bitcoin moving the index. If such firms are a share x of the index and move one for one with Bitcoin, they add $x\sigma^2/\sigma_m^2$ to beta, and at the volatilities of Table 7 Bitcoin's variance is about nine times the index's. On November 10, 2026 the most exposed S&P 500 constituents, Coinbase, Robinhood and Block, were together 0.2% of the index (BlackRock 2026). Strategy and the listed miners are not constituents, and Tesla, 1.65% of the index, held 11,509 bitcoin (Tesla 2026), worth under $1 billion and a negligible share of its value. The three firms add about 0.02 to beta, or under 0.1 point to the required return. At $x = 1\%$, which allows them to move more than four times as much as Bitcoin, the addition is 0.09, or under 0.4 points, well inside the beta range of 1.5–2.0. The shareholders of Bitcoin treasury companies are themselves holders, and are counted among the fund and treasury investors of Section 4.

A.7 How volatility is measured. Every volatility in the paper is the standard deviation of monthly log returns in US dollars, multiplied by $\sqrt{12}$. Two bases are used, and Table 9 shows every asset of Table 7 on both. For the equity indices and gold, Table 7 takes the whole sample, January 2012 to April 2026, measured on the total returns of the iShares MSCI ACWI, Vanguard S&P 500, Invesco QQQ, iShares Russell 2000 and SPDR Gold Shares ETFs; their volatility moves with market conditions but has no trend. For Bitcoin and Ether, whose volatility has fallen with every cycle, the whole sample would describe an asset that no longer exists (87% and 111%), so the paper weights recent months more heavily. With r_{t-k} the monthly log return k months before month t, $\sigma_t^2 = 12\sum_k \omega_k r_{t-k}^2 / \sum_k \omega_k$, with weights $\omega_k = 2^{-k/9}$ that halve every nine months and a sum that runs over every month back to July 2010. There is no fixed window: the latest nine months carry 50% of the weight, the latest eighteen 75% and the latest three years 94%. This gives 42% for Bitcoin in May 2026 and 64% for Ether, and it is the volatility used at every month-end in Section 5. If the equity indices and gold were put on that recent basis, they would be 0.7 to 2.9 points more volatile than in Table 7 and gold 6.1 points more, after its rise of 2025. Their compound returns would then be 0.1 to 0.5 points lower (7.5–7.9% for the four indices) and gold's 1.1 points lower; the comparisons of Section 7 would not change.

The currency matters for one comparison. In dollars the global index is only 0.3 to 0.8 points less volatile than the S&P 500; in euros the gap is 1.2 to 1.8 points, because the dollar tends to rise when equities fall (Appendix E.4). The paper works in dollars throughout. Figures quoted by data vendors can differ from those of Table 9 for two further reasons: they are usually computed from daily prices, and over windows

Table 9: Volatility by window and by currency, % a year.

	2012–2026	Last 5 years	Last 3 years	Last year	Recent-weighted
In US dollars					
Global equities	**13.8**	14.9	12.2	12.7	14.5
S&P 500	**14.1**	15.7	12.9	13.1	15.1
Nasdaq-100	**17.2**	20.4	16.4	18.7	20.1
Russell 2000	**19.1**	20.5	20.0	14.5	19.8
Gold	**15.5**	15.8	16.2	22.0	21.6
Bitcoin	86.9	57.9	45.2	34.7	**43.4**
Ether	111.2	73.1	64.1	62.2	**64.0**
In euros					
Global equities	12.6	13.0	11.2	9.9	12.9
S&P 500	13.8	14.7	13.0	11.1	14.4

Standard deviation of monthly log returns times $\sqrt{12}$; windows end in April 2026, and the first column starts in January 2012. Global equities: the MSCI ACWI. Recent-weighted: squared returns, exponentially weighted with a half-life of nine months. In bold, the basis used in Table 7; for Bitcoin the paper uses the same measure one month later, 41.8% in May 2026, rounded to 42%. Ether's series begins in August 2015. Euro figures convert each monthly return at the month-end exchange rate. Script `41_volatility_table.py`.

that end at a later date.

B. CONCENTRATION RISK

The holder weight w is the input the paper measures least directly, so this appendix shows how it is built (Figure 5). One number is measured: in 2021Q3 US households that owned any crypto held 19% of their financial assets in it on average (Candia, Coibion, Gorodnichenko, and Weber 2026), with the Austrian 2023 survey (OeNB 2025) as a cross-check. Everything else follows from market clearing, capitalization $= \bar{w}\times$ participating wealth, run backward and forward from that anchor.

Why the combined weight. Holders differ in wealth W_i, in risk aversion γ_i and in the weight w_i they choose. Each satisfies the concentration term of equation (1) with her own numbers, $\pi = \gamma_i \sigma^2 w_i$, where π is the premium the asset offers for that risk, so she holds $W_i w_i = W_i \pi/(\gamma_i \sigma^2)$ dollars of it. The holdings must add up to the capitalization C. Summing over holders and solving for the premium,

$$\pi \;=\; \bar{\gamma}\sigma^2 \frac{C}{W}, \qquad \frac{1}{\bar{\gamma}} \;=\; \sum_i \frac{W_i}{W}\frac{1}{\gamma_i}, \qquad W = \sum_i W_i. \tag{10}$$

The premium that clears the market is therefore the concentration term evaluated at the combined weight C/W of equation (2) and at a wealth-weighted (harmonic) average of risk aversions. No single holder sets it. The two readings of Section 5 differ only in whose wealth is counted in W: every owner of crypto, or the early holders alone.

B.1 Participation. The participant count is the share of US adults holding crypto in household surveys: 0.4% (2014) and 1.2% (2018) from the Federal Reserve's payment-choice surveys, 11% in 2021Q3, and 12% in mid-2023 and in 2025 from Candia et al. (2026), whose panel shows a dip of two to three points in between that we do not model. The 2021 level is the least settled. The Federal Reserve's SHED puts adults who used or held crypto at 12% in 2021; the Candia et al. panel reports about 3% in 2021 and 11%

in 2022. We use 11% at the 2021Q3 anchor because the same survey reports portfolio shares as similar in 2021 and 2025, which the back-cast reproduces only if participation is similar too; Appendix D.4 gives the alternative. Other measures differ in level (later SHED waves 7–10%; Pew's broader count 16–17%). The path is interpolated log-linearly between survey dates; the Coin Metrics count of addresses holding a balance supplies the shape only before the first survey point.

B.2 Wealth behind the holders. Participating wealth is anchored at capitalization divided by 0.19 in 2021Q3 (about $4.3 trillion) and moved with the participant count times wealth per participant, a ramp from a quarter of the 2021 level in 2012 to 1.4 times by 2026 (Appendix D.4 replaces the ramp with US net worth per adult). A second class of holders owns Bitcoin through funds and public companies (Table 10). Its first vehicle was the Grayscale Bitcoin Trust, which began in September 2013 and became a spot ETF on January 11, 2024 (Grayscale 2017, 2020, 2021, 2024). Public companies held little before MicroStrategy's first purchase in August 2020 (Akamo 2022; MicroStrategy 2022, 2023). The spot ETFs and the treasury companies then raised the total to 2.5 million BTC by mid-2026 (roughly 1.25 million in US spot ETFs and 1.26 million in corporate treasuries, Strategy alone about 0.85 million), worth about $190 billion at the May 2026 price. Their investors are not "crypto owners" in household surveys and hold small weights, though the evidence on how small is thin. Institutions filing 13F reports held under 1% of their portfolios in Bitcoin ETFs on average in 2025Q3, and accounted for 24% of the funds' assets (CoinShares 2025); among financial advisers' client portfolios with crypto, 64% held more than 2% (Bitwise 2026). We assume 1.5%, at which these investors represent about $13 trillion of participating wealth, against $7 trillion for surveyed owners (panel b of Figure 5). At 3% they represent about $6 trillion, and the lower bound on the 2026 weight is 12% instead of 8%, which raises the central estimate to 19.0% (Table 14). Some fund investors may also count themselves as crypto owners in the surveys; to that extent the two groups overlap and the lower bound is too low. This applies with more force to the Grayscale trust before 2024, many of whose investors were individuals: carrying the 1.5% weight back to them makes the lower edge of Section 5 a lower bound in those years.

Table 10: Bitcoin held through funds and by public companies, million BTC.

	Funds	Companies	Total		Funds	Companies	Total
End of 2013	0.070		0.070	End of 2020	0.607	0.096	0.703
End of 2014	0.135		0.135	End of 2021	0.645	0.216	0.861
End of 2015	0.141		0.141	End of 2022	0.632	0.225	0.857
End of 2016	0.172		0.172	End of 2023	0.620	0.281	0.901
End of 2017	0.176		0.176	Mid-2024			1.15
End of 2018	0.204		0.204	Mid-2025			2.10
End of 2019	0.261	0.020	0.281	Mid-2026	1.25	1.26	2.5

Funds: the Grayscale Bitcoin Trust to 2023, from its filings; US spot ETFs from 2024. Companies in 2022 and 2023: MicroStrategy's reported 132,500 and 189,150 BTC plus the other companies at their 2021 level. The path is linear between these dates and zero before the trust began in September 2013. Halving or doubling the company figures before 2024 moves the lower edge of the band of Section 5 by at most one point and the gaps of Table 5 by 0.1.

B.3 The weight. Equation (3) of Section 5 combines these inputs: C_t is Bitcoin's capitalization at month-end t, q_t the share of adults owning crypto (B.1), k_t wealth per owner relative to 2021 (0.25 in 2012, 1 in 2021 and 1.4 in 2026, linear in between), and I_t the value, at the month-end price, of the Bitcoin of

Table 10. Without I_t the weight counts surveyed owners only; with it the weight also counts fund and treasury investors, as holders with 1.5% of their wealth in Bitcoin. Both are capped at 0.9. They coincide until the trust begins and form a band afterwards: 15–17% in mid-2017, 20–25% in mid-2019 and 12–19% in 2021Q3 (panel d of Figure 1). A worked example: in June 2017 capitalization was 4.9% of its 2021Q3 level, ownership 8.3% of its 2021Q3 level, and wealth per holder two-thirds, so $w = 0.19 \times 0.049/(0.083 \times 0.67) \approx 17\%$. Only C_t is observed monthly; q_t is observed at five survey dates and interpolated, and k_t and the 1.5% are assumptions. The path spikes at the 2013, 2017 and 2021 peaks, because capitalization moves with the price while participation is smooth (Appendix D.3). The 2026 range of Section 4, $w \in [0.08, 0.19]$, runs from the weight with fund and treasury investors to the surveyed value. Two omissions are left as they are. Coins that are permanently lost (estimates of 3–4 million BTC) should be removed from capitalization, which would lower every weight by 15–20%. And the survey average counts each household once, where equation (2) weights by wealth: the combined weight is lower than 19% if wealthier owners hold smaller shares, and higher only if the fifth of owners above half their portfolio hold most of the wealth (Section 8).

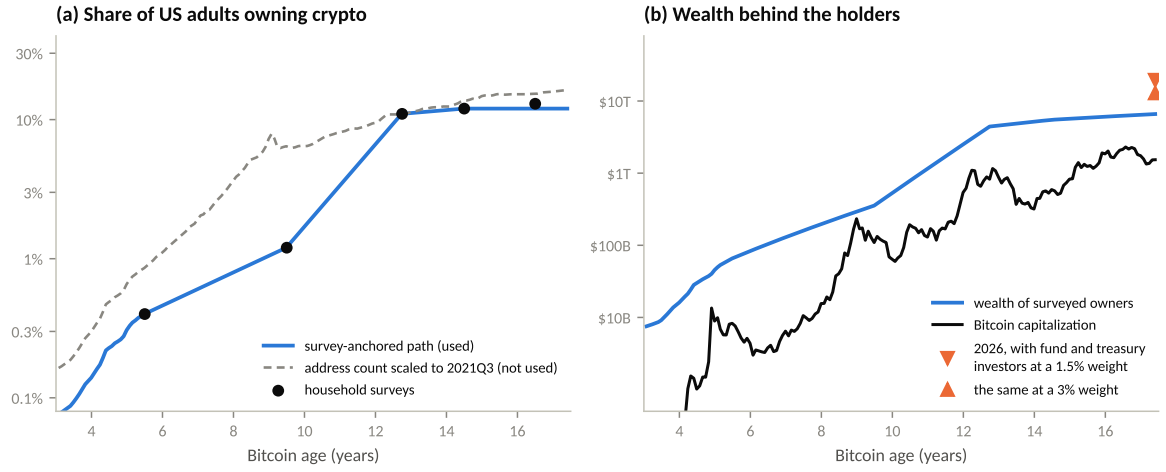


Figure 5: Constructing the holder weight. (a) Share of US adults owning crypto: survey points, the log-linear path used, and the address-count alternative not used. (b) Wealth of surveyed owners, Bitcoin's capitalization, and the 2026 totals once fund and treasury investors are added at weights of 1.5% and 3%. Logarithmic scales. The weight itself is panel (d) of Figure 1.

C. MORTALITY RISK

C.1 Hazard and age. A *hazard rate* answers a simple question: of the assets alive at a given age, what fraction die within the next year? If 100 three-year-old coins are alive in January and 12 are dead by December, the hazard at age three is 12% a year. Estimating hazards across ages, with assets that are still alive treated correctly as "censored" (we know they lived *at least* this long), is standard survival analysis.

The Lindy question. The age rule's most natural economic defense was mortality: perhaps Bitcoin's enormous returns were fair payment for a risk of sudden death that declines as the asset ages. This is the "Lindy effect," under which remaining life expectancy grows with age, so the hazard falls like k divided by age. If that were true with k near 5.6, the age rule would simply be survival compensation, rationally priced, with equation (1)'s mortality term doing all the work.

The census does not support it. Table 11 shows the death rate by age bucket under the curated definition, for the assets the panel tracks; C.4 sets the result against studies of all listed coins.

Table 11: Mortality hazard by age, curated death definition (1,815 assets, 520 deaths, 2010–2026).

Age (years)	Deaths	Exposure (asset-yrs)	Hazard (per yr)
0.1–0.5	6	693	0.9%
0.5–1	37	770	4.8%
1–2	111	1,213	9.2%
2–3	106	905	11.7%
3–4	82	661	12.4%
4–5	78	488	16.0%
5–7	63	502	12.6%
7–9	33	198	16.7%
9–12	4	53	7.5%

Buckets begin at age 0.1 (the first weeks of a listing are definitionally alive) and the sparse 12–17 bucket (7 asset-years, no deaths, Bitcoin the main occupant) is omitted; total exposure across all ages is about 5,640 asset-years.

Among tracked assets the hazard *rises* with age through at least age nine (Figure 6). A Poisson regression of monthly deaths on log age gives an age elasticity of **+0.42 (s.e. 0.04)**, where Lindy requires -1: each doubling of age *raises* the death rate by roughly a third (C.4 has the specification and its variants). A Lindy constant of 5.6 would have implied that a three-year-old cryptoasset had roughly a 1-in-20,000 chance of reaching Bitcoin's current age; the measured elite hazard, held constant, implies a 14-year survival probability around 0.6.

Where the data end. Three caveats belong next to that result. First, in the oldest populated bucket (ages 9–12) the raw hazard falls back to 7.5%, on 4 deaths and 53 asset-years; the rise is established through roughly age nine, the data's support ends near age twelve, and Bitcoin is 17.4. The Lindy rejection is therefore a statement about the asset class over its observed life span, not a measurement of Bitcoin's own current hazard, which is exactly why the pricing below uses the elite hazard *level* rather than an age extrapolation. Second, elite deaths cluster in calendar time (2022, 2024–25), with contagion episodes and delisting purges, not smoothly in age; the three jump-collapses in particular sit within six months of one another and were causally linked (Terra propagated through Celsius to FTX), so the Poisson confidence interval, which assumes independent events, understates the uncertainty. We treat 0.30% as the point estimate of a partly systematic process, which a diversified allocator may rationally charge for in part; Section 7 does so. Third, twelve of the 36 elite deaths fall in 2025, most of them delistings; C.3 gives the hazard without them.

C.2 Size and the elite. Table 1 in Section 3 sorts the 1,806 assets with volume data by peak trading volume into four groups of equal number, and Figure 7 shows the survival curves by size. An asset counts as elite from the month it first ranks in the top twenty (or fifty) by monthly volume, and its exposure is measured from that month. Of the 499 assets that ever reached the top fifty, 118 died, 5.5% a year (95% confidence interval 4.5–6.5%). Most elite deaths are slow fades: OMG, NEM, Bitcoin Gold and their like lost their listings over months or years, during which holders could exit at declining prices; losses of that kind live in an asset's volatility and average return, not in its jump hazard. Four are contested: flagged dead by the drawdown rule although their chains still operate (ICP, SUSHI, FLOW, NANO; C.3). The 520 dead assets are, in the taxonomy of the scarcity framework of Papadogiannis (2025), the "physically scarce but

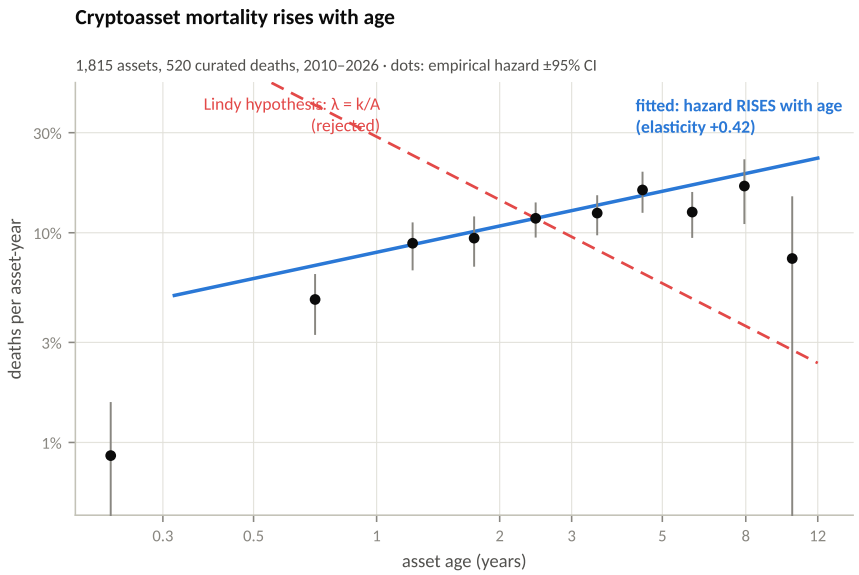


Figure 6: Mortality hazard versus age (log–log). The fitted line uses the period-fixed-effects elasticity of $+0.42$; the dashed line is the Lindy shape $\lambda = k/A$.

no longer desired" quadrant realized at scale: hard-coded supply caps outlived demand, and scarcity without desirability monetizes nothing.

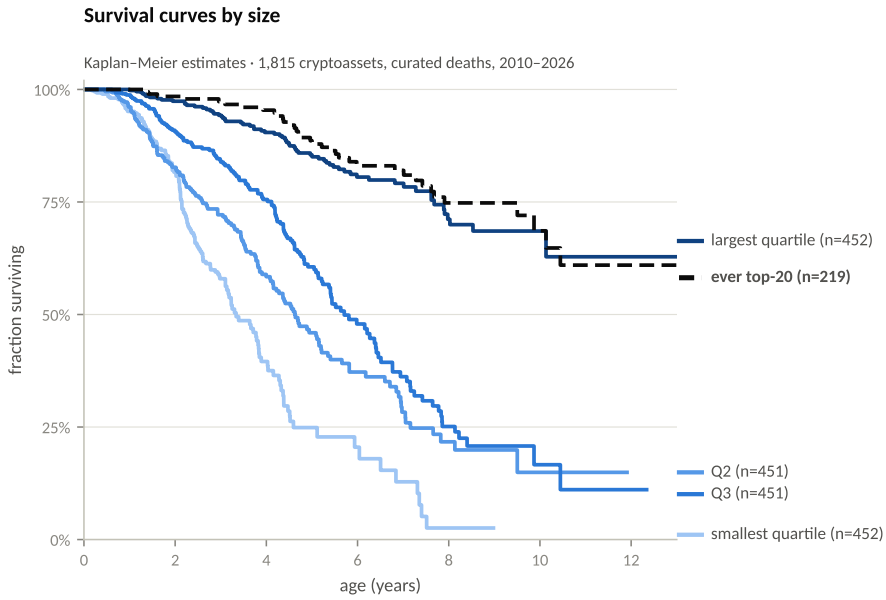


Figure 7: Kaplan–Meier survival curves by peak-volume quartile, with the ever-top-20 cohort (dashed). Half of the smallest quartile is dead by age 3.3; the largest quartile and the elite track each other and remain above 60% survival past age 10.

C.3 The curated elite death table. Table 12 lists all 36 deaths among the 219 ever-top-20 assets, with the classification each received. *Type*: jump = sudden, value-destroying collapse priced as a Poisson event;

exchange = failure of the issuing exchange; fade = gradual abandonment and loss of listings; contested = flagged by a mechanical rule but the chain still operates. δ is the approximate fraction of pre-event value recoverable by a holder (last liquid price or successor claim as a share of pre-event value); it is an estimate, not an accounting figure. *Conf.* flags classification confidence (H/M/L). Dates are the curated economic death dates; mechanical rules alone would misdate LUNA, UST, and FTT, whose collapses are entered from the public record. LUNA and UST are dated May 12, 2022, the day Terra's validators halted the blockchain after LUNA had lost over 99% of its value (Melinek 2022), inside the run of May 7 to 13 documented by Liu, Makarov, and Schoar (2023). FTT is dated November 10, 2022, the day the Securities Commission of The Bahamas froze the assets of FTX Digital Markets and a provisional liquidator was appointed (Securities Commission of The Bahamas 2022); the token had lost three quarters of its value on November 8, and FTX filed for bankruptcy on November 11 (Vidal-Tomás, Briola, and Aste 2023). Dating the three deaths at either end of these windows (May 7 or 13, November 8 or 11) changes no hazard, confidence interval or elasticity reported here (script `40_death_date_check.py`).

Table 12: The 36 deaths among the 219 ever-top-20 assets.

Asset	Death date	Type	δ	Note	Conf.
LTL	2014-12	fade	$\sim$0	minor 2013-era coin	M
FCT (Factom)	2021-07	fade	$\sim$0.05	sponsor company bankrupt 2020; chain stalled	H
LBC (LBRY)	2021-09	fade/regulatory	$\sim$0.05	SEC action 2021; project wound down	H
PPT (Populous)	2021-11	fade	$\sim$0	abandoned	M
ETP (Metaverse)	2022-01	fade	$\sim$0	abandoned	M
NCASH (Nucleus Vision)	2022-01	fade	$\sim$0	delisted majors 2021	H
ZB	2022-02	exchange	$\sim$0	ZB.com halted withdrawals 2022	H
WTC (Waltonchain)	2022-04	fade	$\sim$0	abandoned	H
UST (TerraUSD)	**2022-05-12**	**jump**	$\sim$0.02	algorithmic stablecoin depeg	H
LUNA (Terra)	**2022-05-12**	**jump**	$\sim$0.01	supply hyperinflation; LUNA2 airdrop gave partial recovery	H
QASH (Liquid)	2022-07	exchange	$\sim$0	exchange hacked 2021, absorbed into FTX	H
QC (Qcash)	2022-08	fade	$\sim$0	abandoned	M
FTT (FTX Token)	**2022-11-10**	**jump**	$\sim$0.05	FTX insolvency	H
MITH (Mithril)	2023-03	fade	$\sim$0	serial delistings	H
QSP (Quantstamp)	2023-05	fade	$\sim$0.02	token abandoned	M
SEELE	2023-07	fade	$\sim$0	abandoned	M
VTC (Vertcoin)	2023-08	fade	$\sim$0.02	lost major listings	M
MDA (Moeda)	2023-09	fade	$\sim$0	abandoned	M
BTS (BitShares)	2024-06	fade	$\sim$0.02	community remnant only	M
AE (Aeternity)	2024-10	fade	$\sim$0.02	team wind-down	M
BCN (Bytecoin)	2024-10	fade	$\sim$0	long dormant; delisted majors years earlier	H
OMG (OMG Network)	2025-02	fade	$\sim$0.02	Binance delisting; development ceased	H
KEY (SelfKey)	2025-04	fade	$\sim$0	abandoned	M
LOOM (Loom Network)	2025-05	fade	$\sim$0	abandoned	H
XEM (NEM)	2025-05	fade	$\sim$0.05	major delistings; earlier opt-in XYM airdrop (2021)	H
BTG (Bitcoin Gold)	2025-06	fade	$\sim$0.02	Binance delisting	H
LINA (Linear)	2025-06	fade	$\sim$0	project shut down	M

Table 12: The 36 deaths among the 219 ever-top-20 assets.

Asset	Death date	Type	δ	Note	Conf.
PDA (PlayDapp)	2025-06	fade	$\sim$0.02	post-2024-exploit decline (PLA→PDA swap preceded)	M
YFII (DFI.Money)	2025-07	fade	$\sim$0	abandoned fork	M
FLOW	2025-09	**contested**	n/a	chain operating; >99% below ATH	H
MXC	2025-09	fade	$\sim$0.02	rebranding/decline	L
STMX (StormX)	2025-10	fade	$\sim$0	abandoned	M
NANO	2025-12	**contested**	n/a	active community; lost major listings	M
ICP (Internet Computer)	2026-01	**contested**	n/a	alive, top-100; "death" is an artifact of the listing-day price-spike ATH	H
HIFI	2026-01	fade	$\sim$0	delisted	L
SUSHI (SushiSwap)	2026-03	**contested**	n/a	protocol operating; >99% below ATH	H

Headline sensitivities. All 36 deaths: hazard 3.54%/yr (CI 2.48–4.90). Excluding the four contested cases: 3.15%/yr (CI 2.15–4.44). Treating all twelve 2025 deaths as censoring: 2.36%/yr (CI 1.51–3.51). The mortality term takes the rate of all 36 deaths, 3.5 points, as its upper end; Table 14 shows the estimate with 2.4. Jump-collapses only (LUNA, UST, FTT): 0.30%/yr (CI 0.06–0.86). In the ever-top-50 universe the jump set adds SRM (Serum) and ANC (Anchor), both 2022 casualties of the same two cascades, for 0.23%/yr.

Migrations excluded as deaths (holders received successor value): MATIC→POL, EOS→Vaulta (A), FTM→S (Sonic), GNT→GLM, RNDR→RENDER, BTT redenomination, NU→T (Threshold merger), STRAT→STRAX, MAID→eMAID, HT→HTX, REP→REPv2, AGIX→FET (ASI merger), XZC→FIRO, TOMO→VIC, NPXS→PUNDIX, BNX redenomination, and DGD, whose dissolution refunded the ETH treasury to holders: a death with δ near 1, the mirror image of LUNA.

C.4 The Lindy regression. Table 13 reports Poisson regressions of monthly death indicators on log age, with offset $\log(1/12)$ and standard errors clustered by asset, without and with calendar-year fixed effects (FE), which absorb market-wide waves such as crypto winters. Lindy (hazard $= k/\text{age}$) requires a log-age coefficient of -1. *Earlier survival studies* find the opposite sign for young coins. Among all listed coins about a third are delisted in their first year (Lansky 2020, as summarized by Aldhahi and Alsamaani 2026), and the odds of a deep, lasting drawdown fall by about a sixth with each year of age (Aldhahi and Alsamaani 2026). Our panel differs in two ways. It holds an asset only from the time the data provider adds it, after the first selection has taken place (Section 3). And most of its deaths are abandonments, which take time to observe: a rule based on a fall in volume or price cannot be met in an asset's first months. The census therefore says nothing about the infant mortality of new listings. What it shows is that, among assets significant enough to be tracked, having survived to a given age does not lower the death rate afterwards, which is what a Lindy defense of Bitcoin's returns would need.

Parametric fits agree. A Weibull fit to the curated durations gives a shape of 1.57 (95% CI 1.47–1.67), where a hazard that falls with age needs a value below 1, and a median lifetime of 6.4 years.

C.5 Risks the census cannot see, and the option-implied check. The hazard measured here is a backward-looking frequency, and some risks have never yet fired for anyone. The clearest is quantum

Table 13: Age elasticity of the death rate, by death definition.

Death definition	Deaths	Age elasticity, no FE	Age elasticity, year FE
Strict (feed termination)	234	+0.56** (0.07)	+0.44** (0.06)
Broad (feed + price)	263	+0.69** (0.07)	+0.56** (0.07)
Abandonment (all mechanical)	588	–	+0.51** (0.04)
Curated (final)	**520**	–	+0.42** **(0.04)**
Curated, 2025 feed-wall removed	–	–	+0.65** (0.08)
Curated, birth-cohort FE in place of year FE	520	–	+1.62** (0.10)
Elite (ever top-20) subsample	36	–	+0.68** (0.21)

Standard errors in parentheses, clustered by asset. ** significant at the 1% level; * at the 5% level. Every coefficient differs from the Lindy value of −1 at the 1% level.

computing: a quantum computer able to run Shor's algorithm would break the elliptic-curve signatures that secure Bitcoin. In the equation that is a jump, a λ-event with its own loss fraction, and the loss need not be total: roughly a quarter of the supply sits in addresses with exposed public keys (about 4 million BTC; Deloitte 2019) and is immediately vulnerable, while a telegraphed advance in quantum hardware would allow a migration to post-quantum signatures that preserves most other value. The census cannot price such a risk, but option prices bound it. A put pays its full strike if Bitcoin goes to zero, so its price divided by its strike is an upper limit on the market-implied probability of that event before expiry. On November 12, 2026, with Bitcoin at $73,500, the Deribit put struck at $30,000 and expiring on June 25, 2027 was offered at 0.0035 BTC, about $257 or 0.86% of its strike (Deribit 2026). That caps the probability of a collapse to zero before expiry at 0.9%, or about 1.4% a year (script `35_option_bound.py`). The cap is loose, because most of the put's price pays for ordinary falls below $30,000 and because market-implied probabilities exceed actual ones. It is consistent with the census's 0.30% a year for sudden collapses, and it is the running check on any larger forward-looking hazard, quantum included. An investor who privately assigns, say, a 1% annual probability to a disorderly quantum break with half of value lost should add 0.5 points to the mortality term; the equation accepts the view without modification.

D. ASSEMBLY AND VALIDATION

D.1 Propagating input uncertainty. The 13–26% envelope of Section 4 sums the extremes of four input ranges: beta 1.5–2.0 on a 4% ERP; risk aversion $\gamma \in [1, 2.8]$; holder weight $w \in [0.08, 0.19]$; mortality term 0.3–3.5 points; with the risk-free rate at 5.0% and $\sigma = 0.42$. To obtain a distribution rather than an envelope we draw each input independently and uniformly over its range, 400,000 draws, and compute the required return $r_f + \beta\,\mathrm{ERP} + \gamma\sigma^2 w + \lambda(1-\delta)$ for each. The median is 18.3%, the central 68% range 16.4–20.5%, and the 90% range 15.4–21.9%. These ranges summarize uncertainty about the inputs; they are not statistical confidence intervals. The uniform assumption is deliberately uninformative, and any distribution concentrated toward the interior of the ranges tightens the result. Table 14 changes one assumption at a time. The central estimate stays between 16.6% and 19.8% in every case but one, the low ownership path of Appendix D.4. Independence matters for the width, not the center, and the pair it matters for is risk aversion and the holder weight: if the less risk-averse hold more, the two move oppositely and the range narrows. The same procedure gives Table 6 and the gold figures of Appendix E (beta uniform on $[-0.1, 0.1]$, σ on $[0.13, 0.20]$, w on $[0.02, 0.10]$, mortality on 0–0.3 points: median 5.5%, 90% range 5.0–6.0%) and Bitcoin's

digital-gold endpoint ($\sigma = 0.15$, w on $[0.02, 0.10]$, beta and mortality at Bitcoin's ranges: median 14.2%, 90% range 12.3–16.0%). The script is `23_uncertainty.py`.

Table 14: Sensitivity of the 2026 required return, percent per year.

Case	Median	90% range
Baseline	18.3	15.4–21.9
Mortality term 0.3–1.5 points	17.2	14.8–20.7
Mortality term 0.3–2.4 points (without the top-twenty deaths of 2025)	17.7	15.1–21.2
Mortality term 3.5 points (the high reading alone)	19.8	17.5–23.2
Mortality term 0.3 points (the low reading alone)	16.6	14.3–20.0
Risk aversion 1.7–2.8 (household range)	19.2	16.3–22.4
Volatility 35–50%	18.3	15.3–22.7
Volatility 40–55% (option-implied)	19.4	15.9–24.7
Beta 1.3–2.2 (every window of Table 8)	18.3	15.0–22.2
Beta normal, mean 1.75, standard error 0.45	18.4	14.2–22.9
Holder weight 0.12–0.19	19.0	16.0–22.4
Holder weight 0.08–0.15 (whole asset class, D.4)	17.7	15.1–20.6
Holder weight 0.08–0.12 (combined share below the survey average)	17.2	14.8–19.7
Holder weight 0.04–0.06 (ownership of 3% at the anchor, D.4)	15.7	13.7–17.7
Risk aversion and holder weight move together	18.5	14.9–23.1
Risk aversion and holder weight move oppositely	18.1	16.3–20.0
All four inputs move together	18.4	13.2–25.1

Each row changes one assumption of the baseline and leaves the others as they are. In the last three rows the inputs are drawn with perfect dependence instead of independently.

D.2 The band and the monthly test. *The band.* In symbols, the lower edge of Section 5 is $r_f + 0.85\,\beta\,\mathrm{ERP} + \sigma^2 w_{lo} + 0.3$ points, and the upper edge is the larger of $r_f + 1.15\,\beta\,\mathrm{ERP} + 2.8\,\sigma^2 w_{hi} + 3.5$ points and $r_f + w_t^{\text{early}}\sigma^2$, where w_{lo} and w_{hi} are the average owner's weight with and without fund and treasury investors. The early holder sets the upper edge in 43% of the months. Ending the fully invested era at the 2014 survey (0.4% of adults) instead, the alternative of Section 5, raises the full-sample gap to the upper edge from +7 to +14 points. *The test.* At each month-end t from January 2012 to April 2026 the required return μ_t^{req} is converted to an expected one-month simple return, $e^{\mu_t^{req}/12} - 1$, and subtracted from Bitcoin's simple return over the following month; Table 5 reports twelve times the average difference and its standard error. Converting to a monthly rate before averaging puts the early band averages slightly above a plain average of the monthly values plotted in Figure 2, 157 against 147 for the 2012–14 midpoint and less than half a point after 2020. The standard errors are conventional (172 observations; heteroskedasticity- and autocorrelation-robust errors with six lags give the same t-statistics to within 0.3). The age rule enters in its one-month form, $((A + 1/12)/A)^{5.62} - 1$ at age A. The exponent is held at 5.62 at every date. Re-estimated each month from the daily prices then available, it was 5.5–8.2 over 2012–14, 6.0–6.8 in 2015 and 5.6–6.0 from 2016; with that real-time exponent the age-rule gaps of Table 5 become +41 over the full sample (+48 with the fixed exponent), +121 over 2012–14 (+143), +77 over 2015–17 (+82), and change by two points or less in every period since 2018 (script `37_age_rule_exponent.py`). Returns are heavily right-skewed: the largest month (November 2013, +451%) contributes 31 points to the full-sample mean of 118, and excluding calendar 2013 the delivered average is 69%, the midpoint 57%, and the gaps +50 (lower edge, $t = 2.5$), +12 (midpoint) and −30 (upper edge). *Grouping.* The three-year periods of Table 5 are a

judgment: long enough to average over most of a boom and a bust, short enough to leave five periods. By calendar year Bitcoin's average return lies inside the band in 3 of the 15 years, above it in 7 and below it in 5, with gaps to the midpoint of -281 to $+625$ points and standard errors of 36 to 474: the timing that the required return does not describe (D.3). In two-year periods the midpoint is rejected in 2014–15 (gap -147, $t = -2.9$) and in 2016–17 ($+154$, $t = 2.5$), and in no period since 2018. In four-year periods, about one cycle, the gaps are $+87$, $+64$, $+38$ and $+10$, none significant (script `38_grouping_robustness.py`).

D.3 Level, not timing. The required return passes a test of level and is a poor guide to timing, and the age rule is the better description of the path. Over the 113 overlapping five-year windows starting January 2012 to May 2021, the midpoint at the window's start misses the realized average return by 58 points on average and has a correlation of 0.28 with it; the age rule misses by 28 points, has a correlation of 0.86, and is closer in 90% of windows (32 points, 0.72 and 83% with the exponent re-estimated in real time, as in D.2). The reason is in the inputs. A regression of forward five-year log growth on the concentration quantity $\sigma^2 w$ gives a coefficient of -0.8 ($t = -5.9$; overlapping windows, so the statistic is descriptive): the quantity is highest at market peaks, because capitalization sits in the numerator of w and trailing variance is highest after a run-up, and returns after peaks are low.

D.4 Other paths for the holder weight. Three judgments in the back-cast of Appendix B can be made differently, and Table 15 gives the result of each. *Wealth per owner.* The ramp, a quarter of the 2021 level in 2012 and 1.4 times by 2026, is a judgment about who held Bitcoin. Replacing it with the growth of US household net worth per adult (Federal Reserve Z.1, about $80T in 2013 to $170T in 2026) assumes that early holders were as wealthy as the average American, and lowers the early weights and the lower edge. *Ownership in 2021.* With the path of the Candia et al. panel, 3% in 2021Q3 and 11% by mid-2022 (B.1), the anchor is unchanged but every other weight is lower: 5% in 2017 (baseline 17%) and 4–6% in 2026, at which the 2026 central estimate would be 15.7% (Table 14). We do not use this path, for two reasons. It implies that owners' portfolio share fell by two-thirds between 2021 and 2025, where the survey reports the two as similar. And it gives the owners of 2021 about $560,000 of financial assets each, against $150,000 on the baseline. *The whole asset class.* The back-cast clears Bitcoin's own capitalization against owners' wealth, so it assumes that Bitcoin's share of the asset class stayed at its 2021Q3 value. It did not: the share was 95% in 2013, 38% in mid-2017, 48% in 2021Q3 and 74% in May 2026. Clearing the whole class instead gives a weight of 1.7% in 2013 (baseline 3.4%) and 21% in 2017 (17%). With its 15% in place of 19% as the upper end of w in Section 4, the central estimate falls from 18.3% to 17.7% (90% range 15.1–20.6%). We keep 19% because it is a survey number that the 2025 wave confirms, and because the panel behind this back-cast omits some large assets and so overstates Bitcoin's share today. On every path the gap to the midpoint since 2018 is within one standard error (24).

E. EXTENSIONS

E.1 Gold, as a check on the equation. Gold is the natural comparison, both because Bitcoin is so often called digital gold and because gold is the one other large asset that pays no income, so the whole of its return must come from price. Equation (1) prices it with the same four inputs, every one of which sits at the opposite end of the range from Bitcoin's. Market risk: gold's correlation with equities is close to zero over long samples (Erb and Harvey 2013), so the beta term is a few tenths of a point either way ($\beta \in [-0.1, 0.1]$). In our own sample gold's beta on the S&P 500 is 0.10 (s.e. 0.08) over January 2012 to April 2026 and 0.17

Table 15: Other paths for the holder weight.

	Weight	Band, % a year		Gap to the midpoint	
Path	in 2026, %	2013	2026	2012–26	2018–26
Baseline	8–23	21–231	12–27	+54	+15
Wealth per owner as US net worth per adult	9–29	18–231	12–30	+54	+14
Ownership of 3% in 2021Q3	4–6	15–231	11–19	+61	+21
The whole asset class	8–15	17–231	12–23	+55	+14

Weight: the average owner's, with and without fund and treasury investors. Band: mid-2013 and May 2026. Gap: delivered minus required, as in Table 5. Scripts `28_cryptowide_backcast.py` and `36_ownership_sensitivity.py`.

(s.e. 0.10) since 2020, neither distinguishable from zero; a beta of 0.1 adds 0.4 points. Concentration: gold is the most widely held asset on earth, about 224,000 tonnes above ground, of which 44% is jewelry, 18% central-bank reserves, and 23% private bars, coins, and ETFs (World Gold Council 2026), worth about $29 trillion at today's price, some 6% of global household wealth; its typical holder has a few percent in it. With volatility of 13–20% (13.1% over 1836–2011 in Barro and Misra 2016; higher in the last two years), a holder weight of 2–10%, and $\gamma \in [1, 2.8]$, the concentration term is 0–1.1 points, median 0.3. Mortality: gold has no failure mode that destroys the asset; the closest analogues are sovereign restrictions, such as the United States' compulsory purchase of private gold in 1933 at the old official price before the dollar was devalued, which are jurisdiction risks to a holder rather than deaths of the asset, and we allow 0–0.3 points. Adding the three terms at the middle of their ranges gives the premium: about 0 points for market risk, 0.3 for concentration and 0.15 for mortality, half a point in all. The equation therefore returns a required return for gold today of about 5.5% a year (90% range 5.0–6.0%; envelope 4.6–6.8%; Appendix D.1): the risk-free rate plus half a point, a Sharpe ratio near zero. Gold is also the asset of Table 7 that the equation fits least well, and the figure is best read as an upper bound. Jewelry and central-bank reserves are over 60% of the stock, and their holders receive a service that the equation does not count, the unobserved dividend of Barro and Misra (2016); a holder who values that service requires less from the price. This does not say that gold is safe, or that it will earn 5.5%. Its price moves by about 15% a year. It says that investors cannot expect to be paid for bearing those moves, because they are unrelated to the rest of their wealth and are spread over holders who each own little. The long record agrees, and this is a reproduced result, which is its value here: Barro and Misra (2016) find that gold's real price appreciation averaged 1.1% a year over 1836–2011, about what Treasury bills earned, with a covariance with consumption indistinguishable from zero, and conclude that its expected return should sit close to the risk-free rate. The equation reaches the same place from observable portfolios rather than from a consumption model. Gold's 26% a year from the end of 2023 to November 2026 (LBMA 2026) was news, not a payment for risk, and extrapolating it would be the gold version of the age rule.

E.2 What "digital gold" would mean. If Bitcoin's volatility fell to gold's 15% and its holder base diluted to gold's few percent per holder, the concentration term would all but vanish, as gold's has, and what remained would be the two terms gold does not carry: a beta of 1.5–2.0 and a mortality term of 0.3–3.5 points. The equation then gives about 14% a year (90% range 12–16%; Appendix D.1). Only if those two also converged to gold's, with Bitcoin as uncorrelated with equities as gold and as immune to failure, would its required return reach gold's 5–6%. Bitcoin has been moving away from gold in that respect, not toward

it: integration has lowered its concentration premium and raised its market risk. Since 2020 its monthly returns have a correlation of 0.54 with the S&P 500 and 0.04 with gold (0.16 and -0.02 before), and in a regression on both, the equity coefficient is 2.0 ($t = 5.6$) and the gold coefficient is indistinguishable from zero. Digital gold names a required return near the risk-free rate; Bitcoin's is converging instead on that of a high-beta equity (Appendix A.3).

E.3 Related estimates and approaches. *Asset managers.* Several managers now publish long-term assumptions for Bitcoin. Most are adoption scenarios: a share of a target market is assumed and the price follows, with no discount rate and no risk premium. They run from 3–10% a year over ten years (Galindo, Psaila, and Tailor 2025) and 15% a year to 2050 (Sigel and Bush 2026) to 28% and 30% a year to 2035 (Bitwise 2025; Selby 2025). These are forecasts of price, not required returns, and their spread shows how little an adoption path pins down. J.P. Morgan Asset Management (2026b, pp. 47–48) included Bitcoin in its long-term assumptions for the first time in October 2026: a compound return of 10.0% and an arithmetic return of 20.9%, on a volatility of 55%, with a range of -11.2% to 16.6% compounded. Our 9.5% and 18.3% on 42% are close. The method is different. The 10.0% is a weighted average of three price frameworks (supply growth against money growth, Bitcoin as a complement to gold, and network adoption, with weights of 65, 30 and 5), so it too is a forecast of price; the authors add that Bitcoin "mostly trades like a high beta risk asset" and do not yet include it in their main correlation matrices. *Beliefs and demand.* Benetton and Compiani (2024) estimate the demand for cryptocurrencies from surveys of investors' beliefs, and Duchin, Solomon, Tu, and Wang (2025) ask which beliefs justify a given portfolio weight; a zero weight turns out to need very pessimistic ones. Both go from beliefs to holdings, where equation (1) goes from holdings to the return that justifies them. Milne (2025) finds a zero or negative optimal share for a stock investor when Bitcoin's long-run return is set by efficient-market assumptions and its recent correlation with equities; in our terms a positive share rests on the two premia the CAPM omits (Section 7). *Failure.* Taleb (2021) argues that an asset that pays nothing and has any chance of failure is worth nothing. The pricing of jumps gives a different answer: the holder is paid for the hazard while the asset lives, which is the mortality term. Liu and Tsyvinski (2018) backed a disaster probability of about 0.4% a day out of Bitcoin's returns under risk neutrality; the census measures the frequency directly, 0.3–3.5% a year for assets of Bitcoin's standing. Ammann, Burdorf, Liebi, and Stöckl (2022) measure the survivorship bias that dead coins leave in returns, 62 points a year for equal-weighted portfolios and under one point for value-weighted ones, the cross-sectional counterpart of Section 6. *Undiversified owners.* The concentration term is the cost of capital that corporate finance assigns to owners who cannot diversify: entrepreneurs, whose private equity has earned no more than public equity despite its concentration (Moskowitz and Vissing-Jørgensen 2002), managers who cannot sell their shares (Kahl, Liu, and Longstaff 2003), and venture investors (Kerins, Smith, and Smith 2004). We apply the same term to a traded asset whose holders are concentrated by choice.

E.4 Table 7 for a euro investor. Table 7 is in dollars. Table 16 restates it for an investor whose money is in euros and who does not hedge the currency. Three things change and one does not.

The risk-free rate is the ten-year German Bund yield, 3.4%, against 5.0% in dollars. The yield was 3.6% at the end of September 2026 and 3.3% at the end of October and on November 11–12 (Investing.com 2026; Trading Economics 2026).

The premium of each asset over the risk-free rate is the one of Table 7. A euro investor who hedges the dollar earns the dollar premium on top of the euro rate, because the hedge costs the difference between

the two rates (covered interest parity). One who does not hedge earns the same on average if the dollar is expected to lose that difference, 1.6% a year, against the euro. This is uncovered interest parity, which holds better over long horizons than over short ones (Chinn and Meredith 2004). It is an assumption, and the table rests on it: every average return is 1.6 points lower than in dollars because the euro is expected to gain 1.6% a year.

Volatility is measured on returns converted to euros at month-end exchange rates, over the windows of Table 7. For equities and gold it is lower than in dollars, by 0.3 to 1.2 points, because the dollar tends to move against them: over the sample the euro's monthly change against the dollar has a correlation of 0.2 to 0.4 with their dollar returns. For equities this means that the dollar tends to rise when they fall. It is a tendency and not a rule: the euro fell in eight of the fourteen months in which the S&P 500 lost more than 5%, by 1.0% on average, and the correlation is about zero over the sixteen months since January 2025, when the dollar weakened while US equities fell in March 2025. In a downturn a dollar investor therefore loses twice on shares outside the United States, on their prices and on their currencies, which offsets most of what a global index gains from diversification: backed out of the two indices at a US weight of 60%, the rest of the world has a volatility of about 15% in dollars and 13% in euros. A euro investor is in the opposite position. The dollar's rise cushions her losses on US shares, and her euro-area holdings carry no currency effect. For Bitcoin and Ether the volatility is the same in both currencies. The euro's volatility against the dollar, 7% a year, is small beside theirs, and its correlation with Bitcoin's return is 0.05.

The drag and the compound return follow from the first three as in Table 7.

Table 16: Average and compound required returns in 2026, for a euro investor.

		Required return and risk, % a year, in euros				Sharpe ratio	
	Model	Average	Volatility	Drag	Compound	Average	Compound
Global equities (MSCI ACWI)	CAPM	7.2	12.6	0.8	6.4	0.30	0.24
S&P 500	CAPM	7.4	13.8	1.0	6.4	0.29	0.22
Nasdaq-100	CAPM	7.9	16.9	1.4	6.5	0.27	0.18
Russell 2000	CAPM	8.1	18.8	1.8	6.3	0.25	0.16
Gold	MCM	3.9	14.5	1.1	2.8	0.03	−0.04
Bitcoin	MCM	16.7	42	8.8	7.9	0.32	0.11
Ether	MCM	21.8	64	20.5	1.3	0.29	−0.03

Required returns in nominal euros for an unhedged investor: the 3.4% euro risk-free rate plus each asset's premium in Table 7. Volatility: of monthly returns converted to euros, over the windows of Table 7. Drag: half the variance. Compound: average less drag. Sharpe ratio: the average or compound return, less 3.4%, divided by volatility. A hedged euro investor has the averages of this table and the volatilities of Table 7. Script `45_euro_table.py`.

The ranking of the assets is that of Table 7, and for Bitcoin nothing changes but the level: 16.7% on average and 7.9% compounded, with the same Sharpe ratios. The euro investor's advantage is in equities. Because the dollar cushions her losses, the global index has a volatility of 12.6% in euros against 13.8% in dollars, and its Sharpe ratio is 0.30 on the average return and 0.24 on the compound return, against 0.28 and 0.21. Measured against that index, Bitcoin's compound return is 1.5 points higher in euros (7.9% against 6.4%) and 1.7 points higher in dollars (9.5% against 7.8%), at more than three times the volatility in both.